

SAPIENT

THE UNDERGRADUATE JOURNAL OF
BIOLOGICAL ANTHROPOLOGY

VOLUME 5 - SPRING 2017



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ACKNOWLEDGMENTS

Sapiens would like to thank the Department of Ecology, Evolution, and Environmental Biology at Columbia University in the City of New York for its support, in particular Department Chair and Professor Kevin Griffin, Department Administrator Lourdes Gautier, and Professor Jill Shapiro.



LETTER FROM THE EDITOR

On behalf of my fellow editors, I am thrilled to present to you the fifth volume of *Sapiens*, the Undergraduate Journal of Biological Anthropology. This journal was created to help cultivate interest and academic pursuit in all fields of biological anthropology with a particular emphasis on Human Variation and Genetics, Evolutionary Theory and History, Primate Behavior and Ecology, and Paleoarcheology and Morphology.

This year, our team has worked on expanding our editorial board to students from all majors and disciplines, inviting anthropology students as well as graphic designers, to join our family. Additionally, we have worked on developing a stronger online and campus presence through our weekly newsletter and social events. Lastly, *Sapiens* has worked heavily on including our authors throughout the editing process to produce a product we can all be proud of. Our editorial board is excited to continue to provide a platform for students and artists to publish scientific research in a field we love.

— Humza Mirza

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The Long and Complicated Relationship between Humans and *Yersinia Pestis*

Sterling Wright, The University of Texas at Austin

Introduction

There is perhaps no other infectious disease in human history that has affected human civilization as much as *Yersinia pestis* (Dobson, 2007). The first historic epidemic attributed to *Y. pestis* dates to AD 547-787, and is commonly referred to as the Plague of Justinian (Dewing, 2007). Procopius, a sixth-century Byzantine historian, wrote that infected people developed buboes across their bodies and often died within days of infection, two characteristics associated with *Y. pestis* (Littman and Littman, 1973; Fears, 2004; Harbeck *et al.*, 2013). The second epidemic began in the 14th century and lasted until the 18th century. The second historic epidemic occurred between 1347-1351 and is commonly referred to as the Black Death, in which 17-to-30-million people died (Bos *et al.*, 2011; Boire *et al.*, 2014), or as much as a third of Europe's population (Raoult *et al.*, 2013). The outbreak affected so many people that the disease had an influence on religion, politics and economics in Eurasia (Dobson, 2007; Bos *et al.*, 2016). Although the pathogen would generally not have the same virulence or incidence rate as it did during the fourteenth century, genetic evidence shows that the disease continued to persist in Europe unto the eighteenth century (Bos *et al.*, 2016).

While genetic research has been able to provide substantial evidence for diagnosis for these two historic plagues, there are still many unexplored questions that research needs to investigate. For instance, scholars are unable to explain how the disease was so virulent when the first plague broke out. Archaeological evidence suggests that the disease had existed in Europe for 5,000 years (Rasmussen *et al.*, 2015). Thus, if a lineage of *Y. pestis* had been percolating throughout Europe for so long, what enabled the disease to spread like wildfire and wreaked havoc on countless communities. Moreover, little is known about what ended the first two epidemics, since antibiotics and modern medicine were not available. Furthermore, little research has been done on how the changes of vectors or the variation in immune responses within a population would have impacted the virulence of the disease (Armstrong *et al.*, 2005). In this discussion I will review how advances in biotechnologies, most notably next-generation sequencing (NGS), have spurred unprecedented progress in the understanding of the evolutionary history of *Y. pestis* and what steps scholars should take moving forward.

The Importance of Genetic Research on *Y. pestis*

The body of evidence for the plague includes both historical documents and genetic evidence. For both the Plague of Justinian and the Black Death, scholars have traditionally relied on the descriptions of symptoms provided by authors, such as Procopius of Caesarea, Jean Gethin, Giovanni Boccaccio, and Samuel Pepys, and other eyewitness testimonies to identify *Y. pestis* as the disease underlying the plagues. In order to evaluate how the symptoms of *Y. pestis* have been referenced for diagnosis for the first epidemics, we will look at the symptoms that scholars have historically and scientifically attributed to *Y. pestis*. The writings of Procopius indicate that some victims of the Plague of Justinian died from fever and buboes (swollen lymph nodes), while others died from vomiting and choking on blood (Procopius). Jean

Gethin wrote that a bubo formed in his armpit, which caused him major discomfort and probably led to his death (Cohn, 2008).

Currently there are 11 species that comprise the *Yersinia* genus (Johnson *et al.*, 2015). However, only three of those species are able to infect humans: *Y. pestis*, *Y. enterocolitica* and *Y. pseudotuberculosis*. Whole-genome research suggests that all lineages of *Y. pestis* are descended from an ancient *Y. pseudotuberculosis* lineage (Achtman *et al.*, 1999; Chain *et al.*, 2004; Cui *et al.*, 2013; Johnson *et al.*, 2015). Even though the two species are genetically similar, there are noteworthy differences in their virulence and pathogenesis.

Unlike *Y. pestis*, *Y. pseudotuberculosis* generally leads to gastroenteritis (Jani, 2015). A person contracts *Y. pseudotuberculosis* by consuming infected food or water. *Y. pestis*, on the other hand, is transmitted either by an arthropod vector or by cough droplets, depending on its form. *Y. pestis* has three different pathogenesis forms. Its bubonic form is the most common and is generally transmitted subcutaneously to a human host through the bite of an infected flea (Gage *et al.*, 1996).¹ A flea becomes infected by consuming the blood of an infected rodent that is the original host to the pathogen. The infected flea then transmits the disease to a human host during a second blood meal, enabling the pathogen to enter the human's blood-stream. The pathogen reproduces and within days after infection an individual will experience the formation of buboes (swollen lymph nodes) across the surface of his or her skin, while the disease simultaneously spreads via bloodstream to other organs, such as the liver and spleen where it causes further damage to tissues. The other two forms of *Y. pestis*, septicemic and pneumonic, are less common. Like bubonic plague, the transmission of septicemic plague occurs via the bite of an infected flea. Symptoms include fever and chills, abdominal pain, diarrhea and vomiting, bleeding from the mouth, and blackening of tissues. However, the main symptom that distinguishes the septicemic form from the bubonic form is uncontrollable bleeding, which often results in death within 24 hours. Finally, the last form of plague is its pneumonic form. Unlike the previous forms mentioned, this form does not require an arthropod-vector for transmission. Instead, it is communicable through cough droplets. Moreover, its pathogenesis is attacking the lungs by clogging them with fluids, which makes it impossible to breathe. Like septicemic plague, pneumonic plague can cause death in as little as 24 hours. According to Zimmler and his colleagues, these different deadly communicable routes along with the mention of buboes in historical documents make *Y. pestis* the most likely

1 This difference in transmission between *Y. pseudotuberculosis* and *Y. pestis* is due to *Y. pestis*' acquisition of the *pla* gene (Pouillot *et al.*, 2005; Sun *et al.*, 2014). The *pla* is a cell-surface protease that causes fibrinolysis, the enzymatic breakdown of the fibrin in blood clots, and degrades extracellular matrix and basement membranes (Sebbane *et al.*, 2006). This function enables the disease to inhibit the host's ability to contain the bacteria at peripheral infection sites. Two other mutations, the pMT1 and pPCP1, help the bacterium survive in the bloodstream of fleas and increase its survival in the bloodstreams of human and other mammal hosts (Sebbane *et al.*, 2006; Rajanna *et al.*, 2010).

candidate for the Plague of Justinian and the Black Death (Zimblet *et al.*, 2015).²

Some scholars contend that diagnoses of historical epidemics based on texts, however, are not substantial enough (Raoult, 2003; Duncan and Scott, 2005; Dobson, 2007; Altschuler and Kariuki, 2009). The biggest concern is that multiple diseases can cause similar symptoms (Altschuler and Kariuki, 2009). Altschuler and Kariuki (2009) argue that the flu virus that caused the 1918 outbreak had the potential to cause the two plague epidemics. In addition, Dobson (2007) points out that many authors of the Plague of Justinian and the Black Death fail to mention the presence of dead rats or fleas. For instance, she refers to the writings of Giovanni Boccaccio, an eyewitness to the Black Death, Samuel Pepys, an eyewitness to the Great Plague in London, and Procopius, noting that none of them had any references to rats or fleas (Dobson, 2007). Duncan and Scott (2005) argue that *Y. pestis* was not responsible for the two plagues because many of the historical sources mention that the disease was transmitted person-to-person. They also note that outbreaks of plague were reported in places where no rats or fleas were present and the fatality rate was much higher than it is among modern populations. However, Zimblet *et al.* (2015) demonstrated that a single amino-acid change could enable *Y. pestis* go from vector-borne to droplet-borne, making it possible that an evolved form of the disease allowed it to spread over vast distances in a short period of time. While these authors have brought forth valuable points, genetic research based on both PCR and NGS techniques has suggested that *Y. pestis* was the disease responsible for both the Plague of Justinian and the Black Death (Bos *et al.*, 2011; Raoult *et al.*, 2013; Wagner *et al.*, 2014a; Bos *et al.*, 2016).

In 1998, Drancourt *et al.* (1998) were the first to use PCR identification on ancient skeletons, which were dated to sixteenth century, during the second epidemic. The skeletons in the study were excavated from two graves in France. The first grave was in Lambec and contained 133 skeletons belonging to individuals who were buried between May and September 1590. The second grave contained roughly 200 skeletons belonging to individuals who were buried in May 1722 in Marseille. Their DNA tests indicated that *Y. pestis* was present in France during both the 16th and 18th centuries, giving credence to the idea that *Y. pestis* had been the causative agent for the second epidemic that began in the 14th century. Raoult *et al.* (2000) also used PCR identification, which yielded positive results for *Y. pestis* from skeletons excavated from medieval graves in southern France, providing further evidence that *Y. pestis* was the etiological agent for the Black Death. However, their findings were challenged when other scholars applied the same methods to skeletal remains from other archaeological sites, with negative results (Wood and DeWitte-Aviña, 2003; Gilbert *et al.*, 2004; Prentice *et al.*, 2004). Furthermore, Gilbert *et al.* (2004) conducted a PCR identification test using the DNA sequences for the same genes on 108 teeth from 61 individuals who were buried at five separate medieval archaeological sites in central and Northern Europe. They were not able to amplify any *Y. pestis* DNA from any of the individuals. Therefore, these scholars questioned whether the results from the two previous studies were credible. Since later scholars were unable to reproduce the results, many scholars questioned whether PCR identification of ancient DNA was reliable. However, advancements in genetic testing, most notably the advent of NGS technologies and more precise library preparation methods, have overcome many of the challenges of studying ancient DNA (aDNA).

The maturation of NGS technologies has enabled scholars to increase their understanding of the evolution of diseases to a level far beyond what was conceivable with PCR techniques (Harkins and Stone,

2015). Perhaps the most important contribution of NGS was the ability to provide a more complete DNA profile, allowing scholars to answer more complex questions than just whether *Y. pestis* was present. While PCR is helpful in determining whether a fragment of a bacterial genome is present, NGS has enabled scholars to study the entire genome of ancient pathogens. Sequencing their entire genomes has allowed scholars to ask questions about the migration and evolution of ancient pathogens. In addition to better sequencing technologies, advances in overcoming database bias and better contamination controls have given scholars more robust findings in regards to studying ancient pathogens (Warinner *et al.*, 2015). The following two sections delve deeper into how NGS-based studies have not only revived aDNA research but also cast light on many of the mysteries still shrouding the Plague of Justinian and the Black Death.

The Plague of Justinian

The Plague of Justinian has traditionally been recognized as the first historically-documented epidemic caused by *Y. pestis* (Boire *et al.*, 2014; Wagner *et al.*, 2014a; Rasmussen *et al.*, 2015). According to Procopius, the plague started in Pelusium, in the Nile Delta in Egypt, around c. 450 A.D. (Boire *et al.*, 2014). It is believed that the plague originated in rodent reservoirs in Egypt and then, because of trade, reached the Byzantine Empire and other regions by 540 A.D. (Boire *et al.*, 2014). It afflicted populations in Constantinople, North Africa, the Middle East, Italy, Spain, and central Europe, including along the modern-day French-German border. Procopius suggested that approximately 100 million people died from the disease (Russell, 1968). Although we cannot verify the accuracy of his estimate for number of deaths, there does seem to have been a major reduction in the population of Europe at this time that caused the contraction of the Byzantine Roman Empire (Little, 2007).

Even though less genetic research has been carried out for the Plague of Justinian than for either of the latter two plagues (Wagner *et al.*, 2014a), it has already brought forth valuable evidence to show that *Y. pestis* was in Europe during the time of the plague. Wagner *et al.* (2014a) sequenced *Y. pestis* genomes from the skeletal remains of two individuals (A120 and A76) who were recovered from the early medieval cemetery of Aschheim-Bajuwarenring in Aschheim, Bavaria, Germany. The radiocarbon date for A120 was 533 A.D. $\pm$ 90 years, while the radiocarbon date for A76 was 504 A.D. $\pm$ 61 years. Thus, they were well within the time frame for the first plague. After the researchers isolated DNA for *Y. pestis* in these samples and sequenced those genomes, they used a genetic database, BLAST, to place them in a phylogenetic tree consisting of both ancient and modern strains.

The nodal points on their phylogenetic tree for the Plague of Justinian strain and the second epidemic lineages allowed Wagner *et al.* (2014a) to make inferences about how the disease traveled. Their o.ANT1 group is presently established in Central Asia. Their 1.IN1, 1.IN2, and 1.IN3 groups are presently identified in rodent populations in western, central, and southern China, respectively. Their 1.ORI group emerged in southeastern China and spread worldwide, forming the third pandemic that has contributed to the global spread of the disease up to the present. Therefore, Wagner *et al.* (2014a) believe the strains associated with the Plague of Justinian and the Black Plague were descendants from an early Central Asia lineage, and are also the ancestors to some modern Chinese lineages.

Wagner *et al.* (2014a) suggested that their phylogenetic tree indicated that the lineage that caused the Plague of Justinian had become extinct by the second epidemic, because of the strains sequenced in Bos *et al.* (2011). In addition, the genomic data collected by Wagner *et al.* (2014a) provided useful information about the lineages' virulence. Their study identified two SNPs in the Plague of Justinian lineages that contribute

2 Zimblet *et al.* (2015) demonstrated that a single amino-acid alteration within the *pla* gene of *Y. pestis* enabled the disease to go from a vector borne disease to a droplet-borne disease.

to the disease's virulence factors and isolated a genomic region that included several virulence genes associated with bubonic plague, which they thought made the historically suggested mortality rate of 40%, according to Procopius, plausible (Little, 2007). However, their phylogenetic tree suggested that Procopius' claim that the geographic origin of the epidemic was wrong. Their phylogenetic tree suggested that the lineage originated in China and that the disease spread throughout Eurasia because of established trade routes (*i.e.* the Silk Road).

In addition to providing new genetic information about the *Y. pestis* lineages associated with the Plague of Justinian, Wagner *et al.* (2014a) provided a new perspective on the spread of the disease in the second and third pandemics. Since the second epidemic's lineages were positioned at the base of branch I, Wagner *et al.* (2014a) believed that this phylogenetic position was strong evidence to suggest that the second epidemic lineages were the direct ancestors of all lineages associated with the third pandemic. Their explanation for this phylogenetic positioning was that the *Y. pestis* lineages that had traveled from Asia to Europe and Africa during the second epidemic later traveled back to China.

Some scholars do not agree with the inferences made by Wagner *et al.* (2014a). Drancourt and Raoult (2014) pointed out three major flaws in their claims. In a response letter to the editor of *The Lancet*, which published Wagner *et al.*'s (2014a) study, they pointed out that Wagner *et al.*'s (2014a, p. 323) first claim that the disease had traveled along the Nile by boat going from north to south and that "Historical records are consistent with this scenario..." was wrong. Drancourt and Raoult (2014) noted that the only historical document that Wagner *et al.* (2014a) cited for this claim was a 1951 study by the World Health Organization (WHO) (Pollitzer, 1951). Drancourt and Raoult (2014) argued that the route proposed by Wagner *et al.* (2014a) was not likely because the Nile was not navigable from the Mediterranean Sea to Central Africa in antiquity, meaning that it would have been difficult for a large number of people with the disease to reach Central Africa. Drancourt and Raoult (2014) argue that the plague started in upper Egypt, closer to sub-Saharan Africa, and was then spread by travelers on land rather than travelers going against the Nile. Drancourt and Raoult (2014) also rejected Wagner *et al.*'s (2014a) claim that the Plague of Justinian strain went extinct because neither molecular evidence nor written records were substantial enough to support the argument. They postulate that because the number of Plague of Justinian samples is two, the phylogenetic tree constructed by Wagner *et al.* (2014a) is not accurate. More samples dating to the Plague of Justinian, the Black Death and before the modern period need to be sequenced before determining whether the lineage for the Plague of Justinian became extinct. Following these criticisms, Wagner and his colleagues wrote an author's reply to *The Lancet* clarifying their claims (Wagner *et al.*, 2014b). They defended their argument about the possibility that the strain that caused the Plague of Justinian was from China and not Africa by emphasizing that this was only a hypothetical explanation. They also defended their use of the WHO study by claiming that Drancourt and Raoult had misunderstood their original intent. They cited the WHO study only to show that historical documentation suggested that the third plague had originated in China. Furthermore, Wagner *et al.* (2014b) said that their publication clearly stated that their claims were only suggestions and hypotheses and that future genetic research on archaeological remains dating to the Plague of Justinian was needed. While the debate about the origin of the Plague of Justinian continues, Wagner *et al.* (2014b) clearly showed that the disease was present in Europe during the Plague of Justinian, making it at least a viable candidate as the etiological agent for the plague.

The Second Epidemic

After the Plague of Justinian, Europe did not endure any major

epidemic outbreaks for 600 years (Boire *et al.*, 2014). Between the 10th and mid-14th centuries, Europe experienced a 300% population increase, with most of the increase concentrated in urban settings (Perry and Fetherston, 1997). However, this condensed urban setting would serve as a catalyst for the second epidemic. Taking place during the years 1346-1353, the Black Death killed between 17 to 28 million Europeans, resulting in a population decrease of approximately 30-40% (Perry and Fetherston, 1997; Boire *et al.*, 2014; Brubaker, 2015). The Black Death altered human history by reshaping religion, economics and politics. While the Black Death was the pinnacle of the second epidemic, the second epidemic lasted until the 18th century and involved thousands of local plague outbreaks (Schmidt *et al.*, 2015). However, as with the Plague of Justinian, the diagnosis and estimations of the mortality rate of this plague in different populations have mainly relied on historical records. Only in the last decade has NGS research provided scholars with more reliable data about the causative agent for the second epidemic.

In addition to providing more reliable evidence for diagnosis, NGS has aided scholars in understanding how the disease was able to persist in Europe for so long even though it had such a high mortality rate (Bos *et al.*, 2016). Evolutionary theory suggests that individuals who survived the initial epidemic outbreaks developed immunity to the disease, which would have passed down to future generations. This reproductive advantage would in turn then cause the mortality rate for the disease to decrease. In order to explain why this assumption was not the case between the 14th and 18th centuries, scholars have contrived two scenarios (Bos *et al.*, 2016). The first scenario postulates that a single lineage from Asia was introduced to Europe in 1347 via port cities such as Venice and Genoa and simply remained in Europe until the 18th century (Carmichael, 2014). The other scenario is that several lineages were introduced to Europe at different times between the 13th century and 18th centuries (Schmidt *et al.*, 2015). In order to test this hypothesis, Bos *et al.* (2016) conducted a NGS-based study on skeletal remains dating to the 18th century and compared their sequences to ones dating to the 14th century to test which scenario is more probable.

Bos *et al.* (2016) analyzed skeletal material obtained from the Observance (OBS) collection of the Regional Department of Archaeology, the French Ministry of Culture, and the medical faculty of the Université Aix-Marseille. The material had been recovered from a rescue excavation at l'Observance in Marseille, France, which is where the Great Plague of Marseille (1720-1722) occurred. The scholars conducting the excavation believed that the skeletons were plague victims because of historical documentation for the cemetery but as discussed before, diagnosis based on historical documentation is not reliable (Signoli *et al.*, 2002). In the study done by Bos *et al.* (2016), five teeth were subject to NGS techniques after a PCR identification process for *Y. pestis*, and the results were then mapped onto a phylogenetic tree consisting of both modern and ancient strains of *Y. pestis*.

Since a previous NGS study had recovered the genomes of *Y. pestis* strains from two medieval Black Death victims (Bos *et al.*, 2011), this allowed Bos *et al.* (2016) to analyze the evolution of the disease over a four-hundred-year span. Their results indicated that the OBS lineage is most closely related to one of the lineages found in one of the plague victims from London who died in an outbreak that occurred approximately 350 years before the Great Plague of Marseille (Haensch *et al.*, 2010). Since the OBS strains were placed between two contemporary Black Death strains from London, Bos and her colleagues argued that the disease not only persisted but also diversified in Europe. Moreover, since their phylogenetic tree showed no immediate relationship between any of the OBS strains and any modern strains, they argued that the OBS lineage—the lineage of *Y. pestis* that caused the Great Plague of Marseille—had become extinct.

According to Bos *et al.* (2016), a reasonable explanation for the

distinct position of the OBS lineage was that the Black Death lineage became established in a rodent reservoir within Europe, or western Asia, and evolved in isolation over time. Then, the new strain would be responsible for the minor outbreaks that occurred over the centuries. In short, this contradicted the hypothetical scenario by Wagner *et al.* (2014a) that postulated that the disease had traveled back to China during the second epidemic and then came back to Europe. Since Provence was an active port city that served as an entry point into Western Europe from several different regions, Bos *et al.* (2016) could not suggest a specific geographic location for the rodent reservoir that produced the Great Plague of Marseille.

The geographic origin for the Plague of Justinian, the Great Plague of London and the Great Plague of Marseille may be more complicated than scholars currently think. Rasmussen *et al.* (2015) conducted genetic and chemical research on skeletons dating 5,000 years ago to the Bronze Age. Positive results confirmed that *Y. pestis* was present in Eurasia during the Bronze Age, predating both historically documented epidemics. In addition, since they used NGS techniques rather than PCR identification, they were able to compare genetic differences among the lineages of *Y. pestis*. They discovered that their Bronze Age samples had the threonine mutation at amino acid 259, which is an essential mutation in order for the disease to take on its bubonic form but is not needed for the other two forms (Zimblet *et al.*, 2015). Furthermore, virulence genes associated with the pneumonic and septicemic forms were absent, providing substantial evidence that only the bubonic form was present in Europe during this time.

The findings of Rasmussen *et al.* (2015) reveal that *Y. pestis* had been in Europe for thousands of years. This raises the question of the prevalence of the disease over this period. This NGS research enabled the scholars to push the emergence of the disease back by approximately 3,000 years and should make scholars question whether the lineages of the first two epidemics could have originated from Europe rather than from China. Future research on these lineages of *Y. pestis* and comparing them with ones in China, along the Silk Road, and in Egypt could elucidate the geographic origin of the first two plagues along with the understanding of how the degree of virulence of *Y. pestis* evolved over the millennia.

Elucidating the Coevolutionary Relationship between Humans and *Y. pestis*

An impetus for the field of ancient pathogen genomics is the need to describe how human microbiomes have evolved over time. The growing interest in understanding of the history of infectious diseases has extended beyond ancient health. There are burgeoning concerns about modern versions of *Y. pestis* and other infectious diseases like tuberculosis evolving and potentially becoming resistant to antibiotic drugs (Hoiby *et al.*, 2010). One prominent example of how ancient DNA research has led to the discovery of new routes to fight modern lineages of pathogens is with victims of the Spanish flu, the H1N1 virus, on the Island of Spitzbergen, Norway (Annan *et al.*, 2000; Duncan, 2003). In a study of these remains, scholars pursued the DNA and RNA of the Spanish flu virus so that they could produce antibodies in the case of a future breakout with a similar virulence. Therefore, a similar approach should be adopted by researchers interested in *Y. pestis*. First, since the disease is present in populations across the modern world (Hart and Kariuki, 1998), it is imperative to understand what makes *Y. pestis* more or less virulent. Second, since it is known that a few mutations can dramatically alter its virulence (Chouikha and Hinnebusch, 2012; Sun *et al.*, 2014; Zimblet *et al.*, 2015), this sort of research could help mitigate future outbreaks if the disease were used as a biological weapon (Inglesby *et al.*, 2000) or if a novel lineage acquired resistance to current antibiotics (Galimand *et al.*, 2006).

In order to gain a better understanding of *Homo sapiens*' coevolutionary relationship with *Y. pestis*, future researchers should consider sequencing the human genomes of where the *Y. pestis* DNA was recovered. I suggest that scholars should look for a genetic variant related to blood-type. The frequency of the O blood-type is significantly lower in Europe and Asia than in other parts of the world (Vogel and Motulsky, 2013). While the O allele is the most prevalent in the world (about 63%), in Europe its prevalence is less than 50% (Mourant, 1954). One reason for this low frequency may be due to the bottle-necking effects imposed by the *Y. pestis* epidemics. Research has shown that the antigen of *Y. pestis* is similar to the H antigen found on O-group red-blood cells (Pettenkofer and Bickerich, 1960; Vogel *et al.*, 1960). This similarity means that the plague pathogen is not detected as a foreign substance by the bodies of individuals with blood-type O. Therefore, it is possible that the low frequency of blood-type O individuals in Europe today is due to the fact that blood-type O Europeans during these two plagues were at a selective disadvantage compared to individuals of other blood-types. Conceivably, blood-type O individuals would have been at a much higher risk of dying because their bodies would have less likely been able to detect and combat the *Y. pestis* bacterium (Mielke *et al.*, 2011). To the author's knowledge, no ancient skeletons associated with either of the two pandemics have been tested for their blood-types. The analysis of blood-types using aDNA has already been done in numerous studies (Candela, 1936; Thieme *et al.*, 1956; Micle *et al.*, 1977; Hummel *et al.*, 2002; Haak *et al.*, 2004; Halverson and Bolnick, 2008) so testing for the blood-types of ancient human skeletal material should not be a difficult measure to add.

To gain a better understanding of all of the microbes that were present during the first and second plague epidemics, future studies should also consider extracting DNA from dental calculus of ancient skeletons. Dental calculus is the precipitate that is formed on the surface of teeth by the mineralization of saliva and gingival crevicular fluid (GCF). The precipitate is found in individuals with poor dental hygiene (Warinner *et al.*, 2015). Since studies have shown that calculus yields more DNA than dentine of the same tooth (Warinner *et al.*, 2014) and often has DNA from a number of microorganisms, scientists should consider applying these methods to plague victims.³ It will likely widen the breadth of information about the two epidemics and may even show how other diseases contributed to the two epidemics and aid scholars in resolving some of the inconsistencies between ancient texts and current molecular evidence. A modern example of how two diseases can work in conjunction illustrates this scenario. In 2015, the WHO reported that tuberculosis (TB) was the leading killer of HIV-positive people, with 1 in 3 HIV deaths attributable to TB (Tuberculosis fact sheet ... [updated 2017]). This example shows how diseases do not work in isolation and debunks the "one disease, one epidemic" hypothesis often embraced by scholars interested in historical epidemics. The lack of any genetic evidence for these other diseases in previously-studied samples may be due to current laboratory methods, which typically do not involve investigating multiple microbes.

As technology advances, so does our understanding of the coevolutionary history of humans and diseases. Yet there is still much to be discovered, especially in the case of *Y. pestis* and its relation to the Plague of Justinian and the Black Death. Although studies of the genomes of *Y. pestis* dating to the Plague of Justinian and Black Death have helped us to understand the biological and demographic consequences of historic epidemics, there is still one question that scholars have yet to answer: what specific context triggered each epidemic. While an increase in

³ Warinner *et al.*'s (2014) findings showed that as much as 437 ng mg⁻¹ of DNA were recovered of calculus compared to 0.6 ng mg⁻¹ of DNA were recovered from dentine of the same tooth.

population and trade during the two time periods were factors, neither of these—alone or in combination—offers a complete answer. This lingering question is becoming increasingly important as rising globalization and human migration facilitate the spread of infectious diseases. This sort of research may provide useful information that could prevent a fourth epidemic.

References

- Achtman M, Zurth K, Morelli G, Torrea G, Guiyoule A, Carniel E. 1999. *Yersinia pestis*, the cause of plague, is a recently emerged clone of *Yersinia pseudotuberculosis*. *Proceedings of the National Academy of Sciences* 96,24(November):14043–14048.
- Altschuler EL, Kariuki YM. 2009. Was the Justinian Plague caused by the 1918 flu virus? *Medical hypotheses* 72,2(February):234.
- Annan AP, Berdal BP, Smith CR, Heginbottom JA, Davis JL, Skehel JJ, Oxford JS, Duncan KE, Roberts N, Lewin PK. 2000. Ground penetrating radar surveys to locate 1918 Spanish flu victims in permafrost. *Journal of Forensic Science* 45,1(February):68–76.
- Armstrong GJ, Brown PJ, Turner B. 2005. Evolutionary, historical and political economic perspectives on health and disease. *Social Science & Medicine* 61,4(August):755–765.
- Boire NA, Riedel VAA, Parrish NM, Riedel S. 2014. Lessons learned from historic plague epidemics: the relevance of an ancient disease in modern times. *Journal of Ancient Diseases & Preventive Remedies* 2,2(August):1000114.
- Bos KI, Herbig A, Sahl J, Waglechner N, Fourment M, Forrest SA, Klunk J, Schuenemann V, Poinar D, Kuch M, Golding B, Dutour O, Keim P, Wagner DM, Holmes EC, Krause J, Poinar HN. 2016. Eighteenth century *Yersinia pestis* genomes reveal the long-term persistence of an historical plague focus. *Elife* 21,5(January):e12994.
- Bos KI, Schuenemann VJ, Golding GB, Burbano HA, Waglechner N, Coombes BK, McPhee JB, DeWitte SN, Meyer M, Schmedes S, Wood J, Earn DJD, Herring DA, Bauer P, Poinar HN, Krause J. 2011. A draft genome of *Yersinia pestis* from victims of the Black Death. *Nature* 478,7408(October):506–510.
- Brubaker RR. 2015. *Yersinia pestis* A2. In: Schwartzman J, Yi-Wei T, Sussman M, Dongyou L, Poxton I, editors. *Molecular Medical Microbiology (Second Edition)*. Boston: Academic Press. p. 1845–1865.
- Candela P. 1936. Blood-group reactions in ancient human skeletons. *American Journal of Physical Anthropology* 21,3(July):429–432.
- Carmichael AG. 2014. Plague persistence in Western Europe: a hypothesis. *The Medieval Globe* 1,1:157–191.
- Chain PSG, Carniel E, Larimer FW, Lamerdin J, Stoutland PO, Regala WM, Georgescu AM, Vergez LM, Land ML, Motin VL, Brubaker RR, Fowler J, Hinnebusch J, Marceau M, Medigue C, Simonet M, Chenal-Francoise V, Souza B, Dacheux D, Elliott JM, Derbise A, Hauser IJ, Garcia E. 2004. Insights into the evolution of *Yersinia pestis* through whole-genome comparison with *Yersinia pseudotuberculosis*. *Proceedings of the National Academy of Science* 101,38(September):13826–13831.
- Chouikha I, Hinnebusch BJ. 2012. *Yersinia*–flea interactions and the evolution of the arthropod-borne transmission route of plague. *Current Opinion in Microbiology* 15,3(June):239–246. doi:10.1016/j.mib.2012.02.003
- Cohn S. 2008. Epidemiology of the Black Death and successive waves of plague. *Medical History* 27:74–100.
- Cui Y, Yu C, Yan Y, Li D, Li Y, Jombart T, Weinert LA, Wang Z, Guo Z, Xu L, Zhang Y, Zheng H, Qin N, Xiao X, Wu M, Wang X, Zhou D, Qi Z, Du Z, Wu H, Yang X, Cao H, Wang H, Wang J, Yao S, Rakin A, Li Y, Falush D, Balloux F, Achtman M, Song Y, Wang J, Yang R. 2013. Historical variations in mutation rate in an epidemic pathogen, *Yersinia pestis*. *Proceedings of the National Academy of Sciences* 110,2(January):577–582.
- Dewing HB. 1979. *Procopius*. Cambridge, MA: Harvard University Press.
- Dobson M. 2007. *Disease: the extraordinary stories behind history's deadliest killers*. Quercus.
- Drancourt M, Aboudharam G, Signoli M, Dutour O, Raoult D. 1998. Detection of 400-year-old *Yersinia pestis* DNA in human dental pulp: an approach to the diagnosis of ancient septicemia. *Proceedings of the National Academy of Sciences* 95,21(October):12637–12640.
- Drancourt M, Raoult D. 2014. *Yersinia pestis* and the three plague pandemics. *The Lancet Infectious Diseases* 14,10(October):918–919.
- Duncan C, Scott S. 2005. What caused the Black Death? *Postgraduate Medical Journal* 81,955:315–320.
- Duncan K. 2003. *Hunting the 1918 flu: one scientist's search for a killer virus*. Toronto: University of Toronto Press, Inc..
- Fears JR. 2004. The plague under Marcus Aurelius and the decline and fall of the Roman Empire. *Infectious disease clinics of North America* 18,1(March):65–77.
- Gage KL, Dennis DT, Tsai TF. 1996. Prevention of plague. *Morbidity and Mortality Weekly Report* 45,RR-14(December):1–15.
- Galimand M, Carniel E, Courvalin P. 2006. Resistance of *Yersinia pestis* to antimicrobial agents. *Antimicrobial agents and chemotherapy* 50, 3233–3236.
- Gilbert MTP, Cuccui J, White W, Lynnerup N, Titball RW, Cooper A, Prentice MB. 2004. Absence of *Yersinia pestis*-specific DNA in human teeth from five European excavations of putative plague victims. *Microbiology* 150(February):341–354.
- Haak W, Burger J, Alt KW. 2004. ABO genotyping by PCR-RFLP and cloning and sequencing. *Anthropologischer Anzeiger* 62,4(December):397–410.
- Haensch S, Bianucci R, Signoli M, Rajerison M, Schultz M, Kacki S, Vermunt M, Weston D, Hurst D, Achtman M, Carniel E, Bramanti B. 2010. Distinct clones of *Yersinia pestis* caused the Black Death. *PLoS Pathogens* 6,10(October):1–8.
- Halverson MS, Bolnick DA. 2008. An ancient DNA test of a founder effect in Native American ABO blood group frequencies. *American Journal of Physical Anthropology* 137,3(November):342–347.
- Harbeck M, Seifert L, Hänsch S, Wagner DM, Birdsell D, Parise KL, Wiechmann I, Grupe G, Thomas A, Keim P, Zöller L, Bramanti B, Riehm JM, Scholz HC. 2013. *Yersinia pestis* DNA from skeletal remains from the 6th Century AD reveals insights into Justinianic Plague. *PLoS Pathogens* 9,5(May):e1003349.
- Harkins KM, Stone AC. 2015. Ancient pathogen genomics: insights into timing and adaptation. *Journal of human evolution* 79(February):137–149.
- Hart C, Kariuki S. 1998. Antimicrobial resistance in developing countries. *British Medical Journal* 317,7159(September):647–650.
- Højby N, Bjarnsholt T, Givskov M, Molin S, Ciofu O. 2010. Antibiotic resistance of bacterial biofilms. *International Journal of Antimicrobial Agents* 35,4(April):322–332.
- Hummel S, Schmidt D, Kahle M, Herrmann B. 2002. ABO blood group genotyping of ancient DNA by PCR-RFLP. *International Journal of Legal Medicine* 116,6(December):327–333.
- Inglesby TV, Dennis DT, Henderson DA, Bartlett JG, Ascher MS, Eitzen E, Fine AD, Friedlander AM, Hauer J, Koerner JF. 2000. Plague as a biological weapon: medical and public health management. *Journal of the American Medical Association* 283,17(May):2281–2290.
- Jani AA. 2015. Pseudotuberculosis (*Yersinia*). *Medscape Drugs & Diseases, Infectious Diseases* [Internet]. [cited 2017 May 6]. Available from <http://emedicine.medscape.com/article/226871-overview>
- Johnson SL, Daligault HE, Davenport KW, Jaissle J, Frey KG, Ladner JT, Broomall SM, Bishop-Lilly KA, Bruce DC, Coyne SR. 2015. Thirty-two complete genome assemblies of nine *Yersinia* species, including *Y. pestis*, *Y. pseudotuberculosis*, and *Y. enterocolitica*. *Genome Announcements* 3,2(March–April):e00148–15.

- Little LK. 2007. Life and afterlife of the first plague pandemic. In: Little LK, editor. *Plague and the End of Antiquity*. New York: Cambridge University Press. p. 3-32.
- Littman RJ, Littman ML. 1973. Galen and the Antonine Plague. *The American Journal of Philology* 94,3:243-55.
- Micle S, Kobilyansky E, Nathan M, Arensburg B, Nathan H. 1977. ABO-typing of ancient skeletons from Israel. *American Journal of Physical Anthropology* 47,1(July):89-91.
- Mielke J, Konigsberg L, Relethford J. 2011. *Human biological variation, 2nd ed.* New York: Oxford University Press.
- Mourant AE. 1954. The distribution of the human blood groups. Oxford: Blackwell Scientific Publications.
- Perry RD, Fetherston JD. 1997. *Yersinia pestis*—etiologic agent of plague. *Clinical Microbiology Reviews* 10,1(January):35-66.
- Pettenkofer H, Bickerich R. 1960. On common antigen properties between the human blood group ABO and the pathogen of diseases dangerous for the community. Preliminary report. *Zentralblatt für Bakteriologie: International Journal of Medical Microbiology* 179(July):433-436.
- Pollitzer R. 1951. Plague studies: I. A summary of the history and a survey of the present distribution of the disease. *Bulletin of the World Health Organization* 4,4:475-533.
- Pouillot F, Derbise A, Kukkonen M, Foulon J, Korhonen TK, Carniel E. 2005. Evaluation of O-antigen inactivation on Pla activity and virulence of *Yersinia pseudotuberculosis* harbouring the pPla plasmid. *Microbiology* 151, Pt 11(November):3759-3768.
- Prentice MB, Gilbert T, Cooper A. 2004. Was the Black Death caused by *Yersinia pestis*? *The Lancet Infectious Diseases* 4,2(February):72.
- Procopius, 2007. History of the Wars, Books I and II. Echo Library.
- Rajanna C, Revazishvili T, Rashid MH, Chubinidze S, Bakanidze L, Tsanova S, Imnadze P, Bishop-Lilly KA, Sozhamannan S, Gibbons HS. 2010. Characterization of pPCP1 plasmids in *Yersinia pestis* strains isolated from the former Soviet Union. *International Journal of Microbiology* 2010:1-9.
- Raoult D. 2003. Was the Black Death yersinial plague? *The Lancet Infectious Diseases* 3,6(June):328.
- Raoult D, Aboudharam G, Crubezy E, Larrouy G, Ludes B, Drancourt M. 2000. Molecular identification by "suicide PCR" of *Yersinia pestis* as the agent of Medieval Black Death. *Proceeding of the National Academy of Sciences* 97,23:12800-12803.
- Raoult D, Mouffok N, Bitam I, Piarroux R, Drancourt M. 2013. Plague: history and contemporary analysis. *Journal of Infection* 66,1(January):18-26.
- Rasmussen S, Allentoft M, Nielsen K, Nielsen R, Kristiansen K, Willerslev E. 2015. Early divergent strains of *Yersinia pestis* in Eurasia 5,000 years ago. *Cell* 163,3(October):571-82.
- Russell JC. 1968. That earlier plague. *Demography* 5,1:174-184.
- Schmid B, Buntgen U, Easterday R, Ginzler C, Walloe L, Bramanti B, Stenseth N. 2015. Climate-driven introduction of the Black Death and successive plague reintroductions into Europe. *Proceedings of the National Academy of Sciences* 112,10(March):3020-25.
- Sebbane F, Jarrett CO, Gardner D, Long D, Hinnebusch BJ. 2006. Role of the *Yersinia pestis* plasminogen activator in the incidence of distinct septicemic and bubonic forms of flea-borne plague. *Proceedings of the National Academy of Sciences* 103,14(April):5526-5530.
- Signoli M, Séguy I, Biraben J-N, Dutour O, Belle P. 2002. Paleodemography and historical demography in the context of an epidemic: plague in Provence in the eighteenth century. *Population (English edition)* 57,6(November-December):829-854.
- Sun Y-C, Jarrett CO, Bosio CF, Hinnebusch BJ. 2014. Retracing the evolutionary path that led to flea-borne transmission of *Yersinia pestis*. *Cell Host & Microbe* 15,5(May):578-586.
- Thieme FP, Otten CM, Sutton HE. 1956. A blood typing of human skull fragments from the Pleistocene. *American Journal of Physical Anthropology* 14,3(September):437-443.
- Tuberculosis fact sheet [Internet]. [updated 2017 March]. World Health Organization; [cited 2017 May 6]. Available from: <http://www.who.int/mediacentre/factsheets/fs104/en/>
- Vogel F, Motulsky AG. 2013. *Vogel and Motulsky's Human Genetics: Problems and Approaches*. Verlag, Berlin, Heidelberg: Springer Science & Business Media.
- Vogel F, Pettenkoler HJ, Hehnbold W. 1960. ABO-Blutgruppen. II. Genhiiufigkeit und epidemische Erkrankungen. *Acta genetica (Basel)* 10: 267-294.
- Wagner D, Klunk J, Harbeck M, Devault A, Waglecher N, Sahl J, Enk J, Birdsell D, Kuch M, Lumibao C, Poinar D, Pearson T, Fourment M, Golding B, Riehm J, Earn D, DeWitte S, Rouillard J-M, Grupe G, Wiechmann I, Bliska J, Keim P, Scholz H, Holmes E, Poinar H. 2014a. *Yersinia pestis* and the Plague of Justinian 541-543 AD: a genomic analysis. *Lancet Infectious Diseases* 14,4(April):319-26.
- Wagner DM, Keim PS, Scholz HC, Holmes EC, Poinar H. 2014b. *Yersinia pestis* and the three plague pandemics—authors' reply. *The Lancet Infectious Diseases* 14,10(October):919.
- Warinner C, Rodrigues JFM, Vyas R, Trachsel C, Shved N, Grossmann J, Radini A, Hancock Y, Tito RY, Fiddyment S, Speller C, Hendy J, Charlton S, Luder HU, Salazar-Garcia DC, Eppler E, Seiler R, Hansen LH, Castruita JAS, Barkow-Oesterreicher S, Teoh KY, Kelstrup CD, Olsen V, Nanni P, Kawai T, Willerslev E, von Mering C, Lewis Jr CM, Collins MJ, Gilbert MTP, Ruhli F, Cappellini E. 2014. Pathogens and host immunity in the ancient human oral cavity. *Nature Genetics* 46,4(April):336-344.
- Warinner C, Speller C, Collins MJ. 2015. A new era in palaeomicrobiology: prospects for ancient dental calculus as a long-term record of the human oral microbiome. *Philosophical Transactions of the Royal Society B* 370,1660(January):20130376.
- Wood J, DeWitte-Aviña S. 2003. Was the Black Death yersinial plague? *The Lancet Infectious Diseases* 3,6(June):327-328.
- Zimble DL, Schroeder JA, Eddy JL, Lathem WW. 2015. Early emergence of *Yersinia pestis* as a severe respiratory pathogen. *Nature Communications* 6,7487(June):1-10.

Cranial Gunshot Wounds: An Overview and Case Study

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Introduction and Background

Deaths due to firearm injury, specifically gunshot wounds [GSWs], are happening more and more frequently (FBI, 2016). Although more people getting shot feels like a very recent phenomenon, physicians, pathologists, forensic anthropologists, and firearms experts have been analyzing and gathering data on GSWs since long before recent events. Much is known about the wounding mechanism and resultant damage of a variety of gunpowder-powered missiles (both to the soft and hard tissues) from modern research beginning in-gusto as a review of World War II and Korean War casualties (USAM and Coates, 1962). This article will focus on wounding effects to the cranium as applied to a case study of an unknown individual in Columbia University's collection with a suspected cranial gunshot wound.

The damage inflicted by GSWs primarily relies on kinetic energy transfer into tissues from the bullet (Di Maio, 1999)¹. Kinetic energy levels are expressed in the formula $KE = \frac{1}{2} MV^2$, where M is the mass of the projectile and V is the velocity of the projectile; considering velocity (V) is squared and mass (M) is not, velocity is the primary component of kinetic energy (USAM and Coates, 1962). This was demonstrated in a US Army Medical Service's and Coates' report (1962) when researchers launched 1/8-inch steel balls at varying degrees of velocity into anesthetized cats. The higher velocity the ball, the greater damage; even damage as great as to be explosive in enclosed spaces such as the cranium (USAM and Coates, 1962). The connection of velocity has been demonstrated in all subsequent literature and experience – when velocity increases, damage increases exponentially.

The explosive effect in enclosed body cavities (such as the intracranial space) is due to the temporary wound cavity, one of the primary components of projectile wounding. When a projectile such as a bullet strikes flesh, the spinning bullet's impact both increases pressure in surrounding tissue and sends pieces away from itself incredibly rapidly, creating a temporary wound cavity (Di Maio, 1999). As Di Maio (1999, p. 119) sums nicely, "It is the combination of the crushed and shredded tissue and the effect of the temporary cavity on tissue adjacent to the bullet path (shearing, compression, and stretching) that determines the final extent of the wound." The temporary cavity is only temporary, of course, as tissue is elastic and rebounds back into position – but fragments of tissue (including bone), fragments of the bullet itself, and bruising and lacerations from the initial cavity stretching all remain within sometimes considerable distance of the initial bullet path (Fig. 1). This remaining damage after the temporary cavity has resolved is called the permanent cavity (USAM and Coates, 1962). Exactly how large the temporary cavity is, and thus how far the damage extends in a permanent cavity, is determined by how much kinetic energy is transferred.

¹ Though much information is available on gunshot wounding effects to soft tissue, Di Maio (1999) contains the most comprehensive information related to the skeletal damaging effects of GSWs; as such, it is cited frequently.

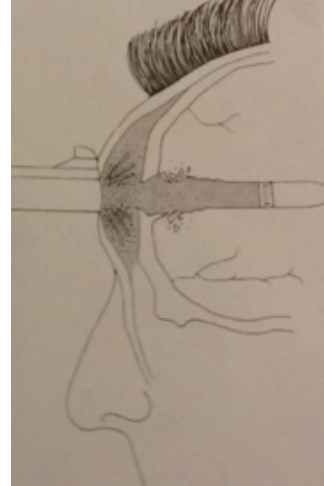


Fig. 1. Wounding effects of a handgun contact cranial gunshot wound (Adapted from Di Maio, 1999)

Therefore, as rifles typically expel bullets at much higher velocities than handguns, they are expected to cause much greater damage upon impact, unless a handgun round hits bone or other immediately vital organs such as the heart (Di Maio, 1999).

The pressure of gasses released from the muzzle of a firearm when fired at very close range is significant (Di Maio, 1999). There are additional wounding effects caused both by the increased pressure and direct damage from the temporary cavity in such cases of contact or near-contact wounds (Di Maio, 1999). "Contact" and "near-contact" are terms used to describe the distance between the muzzle of the firearm and the target – this distance is referred to as "range." There are four ranges of GSWs: contact, of which there are four subcategories of hard-contact, loose contact, angled-contact, and incomplete-contact; near-contact wounds, which can be angled to produce tangential wound defects; intermediate-ranged wounds, which are close enough to leave powder residue but not as close as near-contact; and distant gunshot wounds (Di Maio, 1999). In the latter two damage is typically only done from the bullet itself and the gasses from discharge have little effect (Di Maio, 1999).

When a firearm is discharged, the gunpowder's combustion inside the cartridge case is what propels the bullet down and out of the barrel (James, 2010). The leftover gas is expelled immediately after the bullet in a brief cloud of superheated lead, bullet jacket fragments, and unburned powder that can cause additional injury (Fig. 2). In a contact wound, these gasses are pushed into the actual wound, dramatically increasing pressure and causing explosive damage beyond what only the bullet causes (Di Maio, 1999).

The skull contains both a vital organ and bone and deserves special consideration when discussing the wounding effects of the temporary

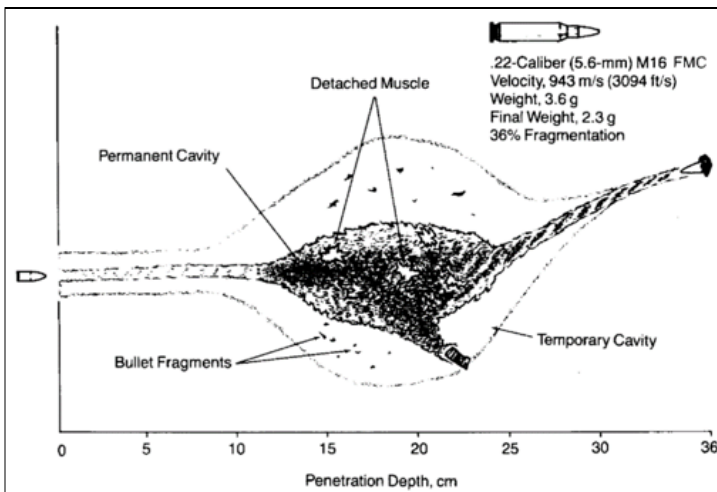


Fig. 2. High-velocity wound track of 5.56mm NATO rifle round (Fackler, 1988)

cavity and contact-range gas pressures. The temporary cavity and gas increase and, with no-where to go, create their own exits in the form of fractures, called “secondary fractures,” with primary fractures radiating from the splitting of the cranial bones at the sites of entry and exit (Fenton *et al.*, 2005). The very thin bone of the orbits including parts of the frontal, sphenoid, ethmoid, and maxillary bones are most susceptible to secondary fractures (Di Maio, 1999), with areas around suture lines being next-most susceptible. As these secondary fractures typically increase as pressure and velocity increase (Fenton *et al.*, 2005), they can be used as a component of narrowing potential weapons used to inflict the injury.

In flat bones, such as the cranial vault, projectiles (bullets) lend an especially useful hand in identifying entry and exit wounds in the form of beveling. A consistent indicator of a GSW to flat bone is internal beveling (for an entry wound) and sometimes external beveling (for an exit wound) (Di Maio, 1999). The direction of this beveling can be used to approximate the direction of the shot (USAM and Coates, 1962; Di Maio, 1999; Fenton *et al.*, 2005), especially in the cases of “keyhole wounds” and perpendicular shots (Harada *et al.*, 2012), although it should be taken with a grain of salt when attempting to assess shot direction when no exit wound or clear bullet path is extant (Quatrehomme and Iscan, 1998a). Nevertheless, the presence of near-symmetrical internal beveling on flat bones is a clear indicator of a bullet wound.

The existing research as briefly introduced above was applied to cranial remains of an individual labeled as No. 150 in the collection of the Physical Anthropology Laboratory at Columbia University in the City of New York. A round frontal bone defect with additional fracturing and missing sections of cranial bone was noted in 150's remains. While it had been posited as a possible gunshot wound, there had been minimal additional assessment regarding confirmation of and proposed information on the defect or the individual's other demographics. This study was conducted to confirm, as much as could be considering the unknown but likely considerable age of the remains, the defect as a GSW and to offer additional information in an attempt to learn more about the history of this individual.

Materials and Methods

All measurements were made using a Neiko Digital caliper with the exception of cranial width, height, and length which were measured

with standard spreading calipers. Sex was assessed using Buikstra and Ubelaker's (1994) “Standards.” Ancestry was assessed following Gill and Rhine (1990). Age was assessed following Ubelaker (1989); however, only dentition was used due to the significant secondary fractures along suture lines and presence of only cranial remains.

Results

Sex: The following scores were assessed from features present on the sample: Nuchal crest = 2 as no “hook” is present and cranial base is flat, although it should be noted that a considerable portion of the cranial base and some of the nuchal area is missing due to exit wound trauma; Mastoid Process = 2; Supraorbital Margin = 3; Supraorbital Ridge/Glabella = 4; Mental Eminence = N/A due to missing mandible. No. 150 presents with additional secondary features as follows: considerably sloping forehead, robust general appearance, large orbits, robust zygomatic arches, minimal cranial bossing. The combination of these features indicates an overall individual rating of 3, which is consistent with departmental seriating and assessment. Secondary characteristics causes this author to lean towards male for this specimen, but officially No. 150 is of indeterminate sex at this time.

Ancestry: A combination of simple cranial sutures, present post-bregmatic depression, unknown incisors, uncrowded dentition, parabolic palate, jagged palatine suture, unknown mandible, unknown chin projection, lesser facial height, little facial prognathism, wide face, retreating zygomatics, s-shaped zygomaxillary suture, rhomboid/slightly rounded, angled orbits, wide interorbital distance, medium nose aperture, tented nasal bridge (although deviated nasal bones indicate healed pre-mortem nasal fracture and complicate assessment), nearly absent nasal sill, and nearly absent nasal spine combine to multiple ancestries. The author is not able to give a definitive ancestry at this time as the above features are evenly distributed across the Asian-European-African-African American spectrum (Gill and Rhine, 1990).

Age: Regarding dentition, there was no mandible with the remains and a total of five maxillary teeth were present: the bilateral M's and P's and a left C. Moderate wear and some chipping of enamel was noted on all five teeth. The left M' presents with a small carie and the C is notably flattened. It is unknown if the enamel chipping occurred pre-mortem or from handling post-mortem. There are signs of pre-mortem eruption of bilateral M's. There is no sign of alveolar resorption, indicating that all maxillary teeth were lost pre- or peri-mortem. Age is assessed as early or middle adulthood (Ubelaker, 1989).

Defect Measurements: There are four major defects (areas of cranial damage) present, three of which were measured and recorded as related to GSW pathology. The fourth, a right maxillary and zygomatic defect, was not measured as initial assessment, lack of wounding characteristics, and distance from proposed wound track indicated post-mortem handling damage. Future research may be performed on this defect as needed. The three other defects were located in the right mid-frontal bone, the right frontal bone in the pterion region, and the occipital bone adjacent to the foramen magnum.

The measurements were as follows: The overall cranial length is 173 mm, width is 138 mm, and height is 142.5 mm. Frontal: the edges of defect are located 10.72 mm laterally from midline and 32.58 mm superior to right supraorbital margin. The defect itself measured 9.27 mm medio-laterally, 10.02 mm superior-anteriorly, and 8.70 mm and 9.62 mm

in opposing oblique measurements. The average was 9.40 mm, or .37 caliber², with a 8.70 mm minimum and 10.02 mm maximum. This defect is uniformly round in appearance and, combined with symmetrical internal beveling, is the entry wound (see discussion, Fig. 3, and Fig. 4).

Right frontal: Location is superior to the speno-frontal suture and anterior to the coronal suture, with each making up an edge of the defect. Of note, the entirety of the internal right orbit—including the sphenoid, ethmoid, frontal, and maxillary sections—is not present. In addition, the speno-temporal and squamous sutures are separated for some distance away from the defect³. The defect presents with external

² A standard decimal measurement of diameter used to measure bullets in the U.S. where 1.0 = one inch (James, 2010).

³ This separation along the suture is present bilaterally but significantly more pronounced on the right side of the skull along the mentioned defect – this increased separation can be attributed to the increase of intercranial pressure and shock waves along the adjacent

beveling and is consists of two sections: a shorter, narrow anterior section and a taller, wider posterior section. The overall width anterior to posterior of the defect is 31.08 mm. The overall height is 24.54 mm. The height of the smaller section of the defect is 9.24 mm and the width of the larger is 19.66 mm. (Fig. 5.)

Occipital: The third measured defect is present on the occipital bone immediately posterior to the foramen magnum; there is no edge of the foramen magnum present at the intersection point. The overall shape resembles two leaves of a four-leaf clover. The defect presents with neutral and external beveling. The furthest point from the former foramen magnum border of the defect is 20.14 mm. The distance from the shorter “leaf” to the former foramen magnum edge is 28.14 mm. The medio-lateral width of the shorter leaf is 23.7 mm and that of the longer leaf is 20.14 mm. The length of the missing foramen magnum edge is 23.7 mm. (Fig. 6).

Discussion

The symmetrical internal beveling and round, “punched out” appearance of the mid-frontal defect indicate that it is an entrance wound from a near-perpendicular angle. Although the exact caliber of the bullet cannot be determined from bone alone (.357 magnum, 9 mm, .380 automatic, .38 Special, etc.), a general caliber class category can be determined from bone, such as the .357-9mm-.38 Special class, which are all between .355 and .357 caliber (Di Maio, 1999; Marshall and Sanow, 2001). The entry wound is uniform with an average caliber measurement of .37, with the width being more narrow than the height. The sub-.355 measured width and above-.355 measured height can be attributed to bone elasticity and the proposed slight downward angle of the shot (Di Maio, 1999). Allowing for variances, the overall measurements still fall well within the range of a caliber studies done with .38 Special (.357 caliber) ammunition (Berryman *et al.*, 1995; Ross, 1996); it is probable that the wound was caused by a round in the .355 to .357 caliber range. The shape characteristics of the wound definitively rule-out an angled shot as there are no indications of furrowing or ‘keyhole-ing’ present when a projectile strikes at a shallow angle (Fig. 7; Di Maio, 1999; Harada *et al.*, 2012; Bonaccorso *et al.*, 2014).

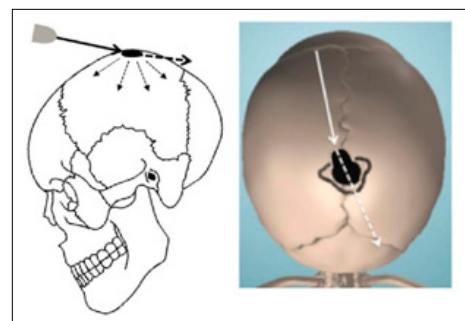


Fig. 7. Example diagram of a keyhole defect (Bonaccorso *et al.*, 2014)

As the entrance wound is perpendicular, insertion of a soft foam rod in a proposed trajectory coincided perfectly with the area encompassed by the occipital defect (Fig. 8). Because the extent of damage is determined significantly by bone thickness (thinner bone is weaker and bullet track (Di Maio, 1999).



Fig. 3. Entrance wound, frontal aspect.

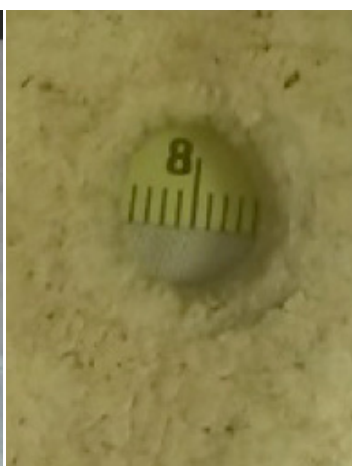


Fig. 4. Endocranial aspect of wound demonstrating beveling effect. Space between dashes are in mm.



Fig. 5. Entrance wound and cranial blow-out defect, right lateral aspect.



Fig. 6. Exit defect and associated damage, inferior aspect.

Figures 3, 4, 5, 6, and 8 by Jonathan Harper

more susceptible to damage) (Quatrehomme and Iscan, 1998a; Di Maio, 1999), there is external beveling present (Di Maio, 1999; Hart, 2005; Spitz, 2006), and the exit is consistently and considerably larger than the entrance with radiating secondary fractures (Quatrehomme and Iscan, 1998b; Di Maio, 1999; Thali *et al.*, 2002; Spitz and Spitz, 2006; Bird and Fleischman, 2015), the occipital defect is consistent with an exit wound.

The right frontal defect and orbital blowout are not in the direct path of the proposed bullet trajectory, nor are they indicative of blunt



Fig. 8. Proposed wound track, right lateral aspect.

trauma as the former has external beveling (Hart, 2005). They are, however, a uniform distance from the proposed wound path: 36.21 mm to the path from the right frontal defect and 29.27 mm from the orbital bones blowout. The location, bones effected, external beveling, and radiating split sutures are all consistent with a contact pistol-caliber gunshot wound (Di Maio, 1999; Hart, 2005). The wounding mechanism at work is as follows: the temporary cavity begins shortly, but not immediately, after bullet entry (which is consistent with the position of the right frontal defect), the orbital plate bones are typically “blown-out” by the increased pressure, and bones are literally pushed out as the gasses try to escape. A centerfire rifle wound can be ruled out, as its much higher generated pressures (even from a distant-range wound) cause massively explosive destruction of the cranium with ejection of the brain and other tissues (Di Maio, 1999). No. 150’s skull is largely intact.

A question remains, though: can the caliber of pistol be narrowed down? The simple answer is that yes, it generally can. There is limited research into the effects of different cartridge pressures with contact and near-contact wounds. The results appear to focus on the extent of entrance-wound radiating fractures and degree of “blow-out.” A study by Fenton *et al.* (2005) included remains of an individual who suffered a contact wound to the mid-frontal bone in 1975 with a “.38 caliber automatic pistol” – exactly like what is suspected for No. 150 (recall that a “.38 caliber” pistol fires bullets of .357 caliber). The wound had extensive radiating fractures that terminated in the exit wound on the occipital (Fig. 9). The dramatic difference in radiating fractures present between Fenton *et al.*’s (2005) remains and No. 150’s remains indicate a difference of cartridge pressure and power. As Fenton *et al.*’s (2005) study did not identify the specific cartridge used, a specific “.38 caliber automatic pistol” cartridge cannot be narrowed down. Comparisons between Fenton *et al.*’s (2005) remains and two other sets of remains from Harada *et al.* (2012) and Bird and Fleischman (2015) can be made, however, and applied to No. 150’s remains.

The Harada *et al.* (2012) and Bird and Fleischman (2015) studies were of contact GSWs with radiating fractures not present at the entry site, but present at exit or along sutures. The GSWs in both studies were

inflicted with low-velocity, low-pressure rounds: .38 Special (Harada *et al.*, 2012) and between a .22 rimfire and .32 automatic (Bird and Fleischman, 2015, Fig. 10). As it was established earlier in this study that increased wound pressure is directly correlated with increased primary and secondary entry wound fractures, the entry wound shown in Fenton *et al.*’s (2005) remains is consistent with a higher-pressure round than those exhibited in Harada *et al.* (2012) and Bird and Fleischman (2015). The proposed entry wound on No. 150 has no radiating primary fractures; thus, the wound exhibited on No. 150 likely falls into the low-velocity, low-pressure round class.

Aspects of No. 150’s past are an enigma – it is of unverifiable ancestry and sex. The age at time of death is known broadly as an adult.

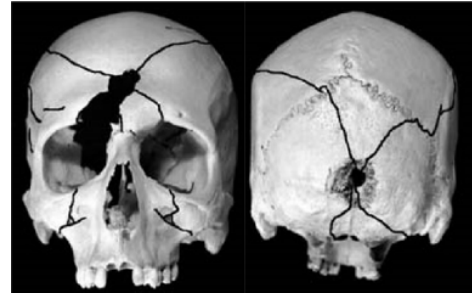


Fig. 9. Example of radial fractures related to contact gunshot wound (Fenton *et al.*, 2005)



Fig. 10. Comparative example of a contact low-pressure GSW to occipital bone with no radial fractures (Bird and Fleischman, 2015)

The circumstances leading up to the gunshot wound are unknown, as is whether it was inflicted pre- or peri-mortem; the cause of death cannot be determined from only a cranium. What is known: indications are No. 150 suffered a contact gunshot wound to their right mid-frontal bone at a perpendicular or near-perpendicular angle that exited just posterior to their foramen magnum. The intracranial pressure from gas expelled from the muzzle of the weapon combined with the temporary cavity created by the bullet itself resulted in secondary fractures to the right orbit and right frontal bone. The causative bullet is consistent with one of .355 to .357 caliber, moving at a low velocity and fired from a low-pressure cartridge. Because of the multiple varieties of firearm ammunition that match these criteria and the unknown time period when No. 150 died, it is impossible to narrow down the exact weapon without specific details on the individual – and even then, it is an unlikely possibility.

Further research needs to be conducted on the variable wounding

effects of gunshots to the cranium. Due to the relatively stable variables and documented ballistic behaviors related to cranial gunshot wounds, (i.e. the skull does not get arthritis and considerably weaken, bone thicknesses are fairly consistent across body types, many pathologies can be readily identified post-mortem, etc.), an opportunity for establishing reliable “standards” of cranial gunshot caliber identification exists. Controversy historically plagues accurate integration of firearm, pathological, and anthropological research (Fackler and Mason, 1996), but the need for continued research has been echoed (Thali *et al.*, 2002; Hart, 2005) with more work still remaining. The remains of the individual used in this study have, in their own small way, contributed to this work.

References

- Berryman HE, Smith OC, Symes SA. 1995. Diameter of cranial gunshot wounds as a function of bullet caliber. *Journal of Forensic Science* 40,5(September):751-754.
- Bird CE, Fleischman JM. 2015. A rare case of an intact bone plug associated with a gunshot exit wound. *Journal of Forensic Science* 60,4(July):1074-1077.
- Buikstra JE, Ubelaker DH. 1994. Standards for data collection from human skeletal remains. *Arkansas Archaeological Survey Research Series* no. 44. Fayetteville: Arkansas Archaeological Survey.
- Bonaccorso L, Gitto L, Maiese A, Dellaquila M, Bolino G. 2014. A keyhole gunshot wound to the head: an autopsy case. *Australian Journal of Forensic Science* 46,3:258-265.
- DiMaio V. 1999. *Gunshot wounds-practical aspects of firearms, ballistics, and forensic techniques*, 2nd edition. New York: Elsevier.
- Fackler ML. 1988. Wound ballistics: a review of common misconceptions. *Journal of the American Medical Association* 259,18(May):2730-2736.
- Fackler ML, Mason RT. 1996. Commentary on Berryman HE, Smith OC, Symes SA. Diameter of cranial gunshot wounds as a function of bullet caliber. *Journal of Forensic Science* 41,4 (July):549-551.
- FBI. 2016. Expanded homicide table 8: murder victims by weapon, 2011-2015. *Uniform Crime Reporting, FBI Criminal Justice Information Services Division* [Internet]. [cited 2017 May 17]. Available from: https://ucr.fbi.gov/crime-in-the-u.s/2015/crime-in-the-u.s.-2015/tables/expanded_homicide_data_table_8_murder_victims_by_weapon_2011-2015.xls
- Fenton TW, Stefan VH, Wood LA, Sauer NJ. 2005. Symmetrical fracturing of the skull from midline contact gunshot wounds: reconstruction of individual death histories from skeletonized human remains. *Journal of Forensic Science* 50,2(March):274-285.
- Gill GW, Rhine, S. 1990. *Skeletal attribution of race*. Alburquerque: Maxwell Museum of Anthropology.
- Harada K, Kuroda R, Nakajima M, Takizawa A, Yoshida KI. 2012. An autopsy case of a decomposed body with keyhole gunshot wound and secondary skull fractures. *Legal Medicine* 14,5(September):255-257.
- Hart GO. 2005. Fracture pattern interpretation in the skull: differentiating blunt force from ballistics trauma using concentric fractures. *Journal of Forensic Science* 50,6(November):1276-1281.
- James CR, editor. 2010. *The ABCs of reloading*. Iola, WI: Krause Publications.
- Marshall EP, Sanow E. 2001. *Stopping power: a practical analysis of the latest handgun ammunition*. Colorado: Paladin Press.
- Quatrehomme G, Iscan MY. 1998a. Analysis of beveling in gunshot entrance wounds. *Forensic Science International* 93,1(April):45-60.
- Quatrehomme G, Iscan MY. 1998b. Gunshot wounds to the skull: comparison of entries and exits. *Forensic Science International* 94,1-2(June):141-146.
- Ross AH. 1996. Caliber estimation from cranial entrance defect measurements. *Journal of Forensic Science* 41,4(July):629-633.
- Spitz W, Spitz D. 2006. *Spitz and Fisher's medicolegal investigation of death*, 4th edition. Springfield, IL: Charles C Thomas.
- Thali MJ, Kneubuehl BP, Zollinger U, Dirnhofer R. 2002. The “skin-skull-brain model”: a new instrument for the study of gunshot effects. *Forensic Science International* 125,2-3(February):178-189.
- Ubelaker, DH. 1989. *Human skeletal remains: excavation, analysis, interpretation*, 2nd edition. Washington, DC: Taraxacum.
- USAM, Coates JB. 1962. *Wound ballistics*. Rockville, MD: Office of the Surgeon General, Department of the Army.

Understanding Mississippian Stone Box Graves: An Osteological Analysis of Three Infants from the Helton Site in the Lower Illinois River Valley

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Introduction

Around 1260 CE at a small settlement known as Helton in the Lower Illinois River Valley (LIV), three infants were interred in boxes constructed of limestone. Two of these children were buried with one sherd each of the same ceramic vessel; the third child was buried with nothing; a fourth stone box lay empty. An analysis of the individuals interred in these stone box graves, contextualized within the mortuary practices of other contemporaneous populations in nearby regions, may shed light onto the possible significance of affording these children were mortuary treatment that was unique to the site and the LIV.

Background

The Mississippian period has eluded a precise definition. Generally, scholars use the term to describe Native American populations living in the Midwest and Southeast from 1000-1500 CE that had complex, ranked social organization, participated in widespread exchange networks, engaged in intensive maize agriculture, constructed large earthen mounds, and adopted a common belief system (Pauketat, 1994). Around 1050 CE, the once modest town of Cahokia exploded into a major polity in an archaeologically visible event known as “the Big Bang.” Located in what is known as the American Bottom, directly across the Mississippi River from St. Louis, Cahokia proper is thought to have sustained upwards of 16,000 people, making it the largest indigenous city north of Mexico by a factor of five (Pauketat, 1998). It has been referred to as the regional “capital” of the American Bottom by some authors and “a new order” by others, highlighting its economic, political, and religious significance both within the American Bottom and beyond until its eventual abandonment around 1300 CE (Farnsworth *et al.*, 1991; Alt, 2001).

The cultural period preceding the Mississippian, known as the Late Woodland period, lasted from 400-1000 CE (Delaney-Rivera, 2004). While the archaeological record of the LIV appears to lack evidence for local adoption of Mississippian lifeways, or “Mississippianization,” following the Late Woodland period, we can assume that at least some communities in the LIV engaged in frequent and continuous contact with the residents of Cahokia and the American Bottom more broadly, given their location in Cahokia’s “inner sphere” at less than 100km away (Pauketat, 1998). Because of this, it is likely that Helton also experienced at least some degree of Mississippianization. Importantly, a recent genetic study conducted by Reynolds *et al.* (2015) confirmed prior beliefs that that Mississippian populations were biologically continuous with Late Woodland peoples in the LIV.

Dating to no earlier than 1250 CE in the American Bottom, stone box graves were a major part of burial practices across the Mississippian southeast for centuries, numbering in the tens of thousands in Tennessee and Kentucky alone (Brown, 1981, Fig. 1). Based on the approximate dates of the Mississippian materials in the LIV, most of which are dated to no later than roughly 1150 CE (Delaney-Rivera, 2004), the absence of stone box graves in the region is therefore not surprising. Rather, their *presence* is anomalous. When compared to the other sites in the LIV, Helton stands out as the sole site containing Mississippian materials

dating to later than 1200 CE, aside from the Starr Village site (Fig. 2). Though it has been established that there was not in fact a Mississippian “void” in the LIV, there remains a near absence of Mississippian sites in the region, as well as a notable scarcity of Mississippian artifacts (Farnsworth and Emerson, 1991).

These observations give rise to numerous questions: Why does the LIV lack evidence for substantial Mississippian contact compared to nearby regions, and why does this contact essentially cease or become archaeologically invisible after 1150 CE? Why were stone box graves all but absent in the LIV, and why did the local community at Helton choose to construct them? Why were HN 46-47 S2-12, HN 46-47 S2-13, and HN 46-47 S2-10 interred in stone box graves, and why was one stone box empty? An analysis of the infants interred in these boxes may shed light onto potential hypotheses.

Helton

Helton is located between Koster Village and Macoupin Creek in Greene County, Illinois. Mounds 46 and 47 are located on a secondary



Fig. 1. Artistic rendering of an individual in a flexed position within a stone box grave (Roe, 2011).

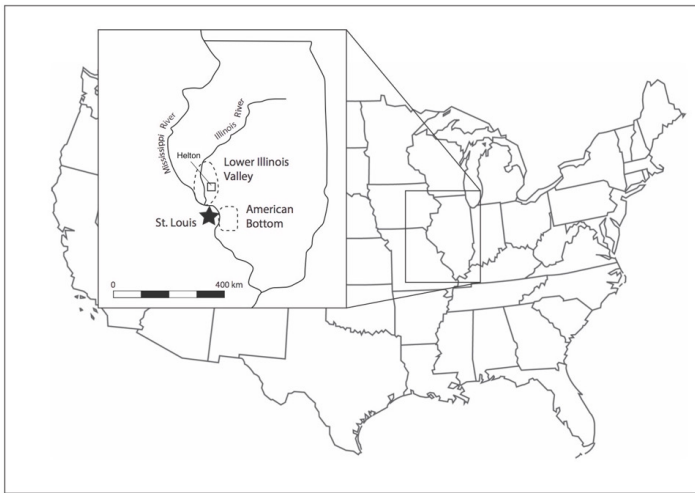


Figure by Amanda Rossillo

Fig. 2. Location of the LIV and Helton within the United States.

ridge in the Macoupin Valley. HN-46 was 15m by 10m and was 1.7m tall, while HN-47 was 17m by 11m and 2m tall. The mounds were excavated from 1974-1978 by Jane E. Buikstra, then of Northwestern University. These mounds were located 5m apart and formed a right angle with each other. This angle contains Stratum 2, which runs from the south end of HN-47 to the east end of HN-46 and contained 17 burials representing 19 individuals located in a Late Woodland cemetery area just south of HN-46 (Conner, 1984). Interestingly, several of these burials were more elaborate than normal mortuary pits, including the three juvenile individuals that were interred in limestone box graves comprising this study. There was also a fourth stone box that lay empty (Conner, 1984). The only additional artifacts associated with the burials were two portions of a single Mississippian shell-tempered jar that were found in two of the stone box graves (Conner, 1984).

Methods

To estimate age at death, a combination of four methods was used when possible. Tooth eruption and root completion were assessed on deciduous and permanent teeth following The Chronology of the Human Dentition Chart by Logan and Kronfeld (1933) as presented in Buikstra (2016), while the degree of root and apex completion of the permanent lower teeth were assessed using the method outlined by Smith (1991). The formation and eruption of the deciduous and permanent teeth and long bone diaphyseal lengths were assessed according to Ubelaker (1978). Taphonomic analyses were performed according to the degree of weathering present following the standards of Behrensmeyer (1978) and Buikstra and Ubelaker (1994). Finally, pathological analyses were undertaken following Buikstra and Ubelaker (1994). The elements were evaluated in terms of shape and size, whether the pathology occurred antemortem or postmortem, the nature of the pathological processes and the degree of healing if any, and whether these processes were primary or secondary.

Results

HN-46-47 S2-12

HN-46-47 S2-12 (Fig. 3) was recovered from its limestone box in a flexed position oriented along the true North-South axis. While the bones were found in the "anatomically predictable positions," the lack of complete articulation suggests that the child was placed in the box after it had decomposed. No other organic or archaeological material was recovered from the box.

Based on the results, HN-46-47 S2-12 is estimated to have been 11-14

months at the time of death. The Chronology of the Human Dentition chart yielded an age range of 11-14 months based on the enamel completion of Ldm², which provided the lower limit of 11 months, and on the fact that Rdm¹ had not erupted, which occurs at 14 months. The Smith (1991) method provided a somewhat higher and wider range of 16-22 months on the basis of crown formation. The crowns of the mandibular first molars were at least half complete, which occurs at 16 months, but were not yet three-fourths complete, which occurs at 22 months. The Ubelaker (1978) method yielded an age estimate of 8-16 months based on the eruption of Ldm¹ and the fact that Ldm² had not yet erupted. Long bone diaphyseal lengths provided consistent results of 6 months to 1.5 years, although the range given by the femoral length was slightly wider, ranging from newborn to 1.5 years of age.

Because dental formation has a strong genetic component and is subsequently less influenced by environmental factors than skeletal elements, age estimates provided by dental analyses tend to be privileged over those yielded by skeletal analyses (Ubelaker, 1987; Scheuer and Black 2004). Long bone diaphyseal length in particular is problematic for numerous reasons (see Ubelaker, 1978). However, because the dental and skeletal estimates provided similar results, they were both factored into the final age at death estimate. If the highest of the lower limits and the lowest of the upper limits of all methods were used as the final age estimate, the resulting range would be 16-14 months. Because all methods except for the Smith (1991) method provided lower bounds that were lower than 16 months, the next highest of the lower bounds was chosen, giving a lower limit of 11 months. This adjusted the age range to 11-14 months, which was considered the overall age at death estimate for this individual.

The frontal bone was classified as Stage 0 with light pitting on the left frontal eminence of the frontal bone and on the left posterior aspect of the squamous portion. Evidence of light root etching was visible as a red staining on the squamous portion. The left parietal was categorized as Stage 1 based on the light pitting dispersed on the anterior and medial aspects near the coronal and sagittal sutures, and root etching present on all aspects. The right parietal and occipital were scored as Stage 0, although traces of dispersed root etching were visible on all aspects of both elements and the right parietal exhibited light pitting along the anterior aspect of the sagittal suture.

The left and right femora demonstrate similar types and quantities of taphonomic activity on all aspects. The distal shafts were classified as Stage 0 and did not appear to be affected by taphonomic activity, though the left diaphysis did bear evidence of light root etching dispersed on the shaft and distal end. The distal ends were also classified as Stage 0, although slight white discoloration and root etching were present. All available aspects of the tibiae were also classified as Stage 0, with some evidence of light pitting on the proximal end and diaphysis of the left tibia.

Though the proximal end of the right fibula was absent, the diaphysis was present and classified as Stage 0. It exhibited slight root etching as evidenced by the white discoloration and sharp edges of the groove, though no other characteristics were noted. The distal end was also evaluated to be Stage 0. It was broken from the shaft postmortem as shown by the lightly colored, jagged edges of the exposed surface of the fracture. There was also postmortem activity inferior to the metaphyseal surface that measured 3.5mm proximodistally on the shaft and was associated with damage to the metaphyseal surface. While approximately half of the proximal end of the left fibula was absent due to postmortem breakage demonstrated by the light, jagged edges, the remaining half was scored as Stage 0.

Based on the visual assessments of taphonomic activity, it does not appear as though this individual was exposed to the elements for a long period of time. Because the bones were found in approximately

correct anatomical positions but disarticulated, it is possible that this individual had received a bundle burial in which the deceased is wrapped in protective material such as cloth and then interred after the soft tissues had disintegrated. This is especially plausible given the frequency of bundle burials in both the Late Woodland and Mississippian cultural traditions (Conner, 1984).

Evidence of cribra orbitalia (Fig. 4) was noted on both orbital surfaces of the frontal bone. Interestingly, no expansion of the diploë was noted. The degree of porosity ranged from very indistinct to true porosity as described by Buikstra and Ubelaker (1994). The edges of the pores appear to be a combination of sharp and smooth, which is indicative of both healed and active periosteal new bone addition. There was also abnormal porosity on both parietals along the sagittal suture and on the occipital bone near the lambdoid suture that was associated with layers of actively healing periosteal new bone formation, as demonstrated by the smooth outlines of the pores and reintegration into the surrounding surface.

Although cribra orbitalia was previously believed to be caused primarily by anemia, it is now understood to be a result of nutritional

deficiencies, infectious disease, and/or parasitism (Buikstra and Ubelaker, 1994; Waldron, 2009). While cribra orbitalia is indeed often clinically associated with anemia, it is usually characterized by an expansion of the diploë due to the need for increased red blood cell production. Because the diploë is filled to capacity with hemopoietic bone marrow, lesions on the cranial vault often form due to increased pressure on the outer table (Lewis, 2006). The orbital surfaces of HN-46-47 S2-12 did not demonstrate expansion of the diploë, nor were the majority of the lesions considered active and/or severe. Therefore, it does not appear as though this individual was suffering from anemia at time of death.

Given the presence of cribra orbitalia and healing periosteal new bone formation at time of death, it is possible that HN-46-47 S2-12 was suffering from a chronic or infectious disease, congenital condition, the consequences of trauma, and/or nutritional stress. Although it is unlikely that this individual was suffering from iron-deficiency anemia, a differential diagnosis cannot be made. Nevertheless, it is clear that this individual was certainly suffering from at least one pathological condition, and that he or she displayed signs of active pathological processes, active healing, and healed lesions.

HN 46-47 S2-13

HN 46-47 S2-13 (Fig. 5) was recovered in a flexed position oriented approximately along the East-West axis. Though most of the elements were in approximately anatomical position, the skeleton was disarticulated which suggests that processing occurred in situ, with certain elements, including most of those of the feet and hands, left clavicle, and left femur, removed. One fragment of a Mississippian ceramic vessel was also recovered from the box, as was one piece of chert and one shell fragment.

Based on the results, HN 46-47 S2-13 was estimated to be 2-3 years of age at time of death. The Chronology of the Human Dentition chart provided an age range of 2-3 years based on the eruption of Rdm² which occurs at 24 months, and the lack of root completion which occurs at 3 years. The incomplete root formation of Rdm² and the enamel formation on RM¹ provided support for this upper bound, as these events would have occurred between 2.5 and 3 years. The Smith (1991) method was unable to be used because LM¹ and RM¹ were obscured by the mandibular bone overlying their respective crypts. The Ubelaker (1978) method yielded an age range of 1.3-2.6 years based on the eruption but lack of root completion of dm¹ and dm². The majority of the long bones were damaged postmortem and therefore the diaphyseal lengths could not be measured. However, the diaphyseal length of the right tibia and the length of the right ilium was able to be estimated, which both yielded age ranges of 1.5-2.5 years. Although the age ranges yielded by all of the methods overlapped, the dental data was privileged in the final estimated age at death because dental age is less variable than skeletal age (Scheuer and Black, 2004). The upper bound of 3 years was based on the degree of root completion, which along with enamel formation is often considered to be the best means to estimate age at death (Ubelaker, 1987). Thus, the age at death for HN 46-47 S2-13 was estimated to be 2-3 years.

Like HN 46-47 S2-12, HN 46-47 S2-13 exhibited minimal evidence of weathering in the form of cracks, though increased pitting was observed on the cranial and long bones. The squamous portion of the frontal bone was scored as Stage 1 due to the concentrated pitting on the anterior, medial, and lateral aspects. The left parietal showed no signs of weathering, though there was postmortem damage on the posterior aspect along 25mm of the lambdoid suture. The right parietal also did not show signs of weathering and was also damaged postmortem into three large fragments. The pitting on the occipital was more moderate and concentrated on the lateral and central aspects, which resulted in a classification of Stage 1. All available aspects of the right femur were

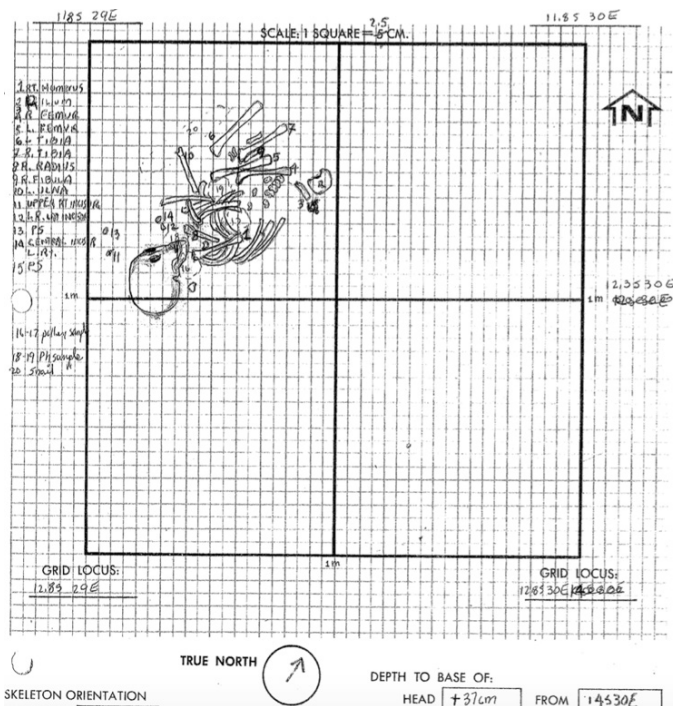


Fig. 3. Schematic diagram of HN 46-47 S2-12 in its burial context (Warren, unpublished data).



Fig. 4. Inactive cribra orbitalia in a juvenile individual from Copper Age Spain (Trancho GJ, n.d.).

scored as Stage 1 due to light pitting and flaking on the anteromedial aspect of the proximal end, and light pitting on all aspects of the diaphysis. The distal end was absent due to postmortem damage as evidenced by the lighter, jagged edges.

Though the majority of the proximal end of the left tibia was damaged postmortem, the diaphysis was classified as Stage 1 based on the light pitting on all aspects, particularly the lateral midshaft. The distal end was considered to be Stage 0, with no evidence of pitting or discoloration. Although there was pitting present on the medial aspect of the proximal end of the right tibia, it was classified as Stage 0 because it was not concentrated in a specific location nor was it associated with cracking or flaking. The tibial shaft was classified as Stage 1 because there was moderate pitting on the anterior, medial, and lateral aspects that was associated with black speckling. There was also postmortem breakage exposing the inner cavity from the anterior distal end to approximately midshaft that measured 47.4mm from the metaphyseal surface from the posterior aspect. In addition, slight root etching was noted on the distal shaft. The distal end was evaluated to be Stage 0, though the anterior aspect was damaged due to the aforementioned breakage. The shaft of the left fibula was the only part of the element present, and was scored as Stage 2 on the basis of concentrated pitting that resulted in flaking of the cortical surface, in addition to root etching.

Like HN 46-47 S2-12, the disarticulated skeleton and excellent preservation of HN 46-47 S2-13 suggest that this individual had not been exposed to the elements to facilitate decomposition. It is likely that the infant had also been given a bundle burial and had undergone further processing, with the removal of elements of the hands and feet occurring in situ.

Abnormal porosity associated with bone addition was noted on both parietals along the sagittal suture. Based on the smoothness of the pores, in addition to the incomplete reintegration of the periosteal new bone into the surrounding surface, it appears as though this individual was actively healing from at least one pathological condition at time of

death. The porosity spanned 61mm of the region immediately adjacent to the sagittal suture of the left parietal and 56mm of the region immediately adjacent to the sagittal suture of the right parietal. No other evidence of pathological activity was noted.

Because periosteal new bone formation may be a response to numerous stressors, including trauma, some congenital conditions, cancers, and overlying soft tissue lesions (Waldron, 2009), the cause(s) of this bone addition and porosity is unknown.

HN 46-47 S2-10

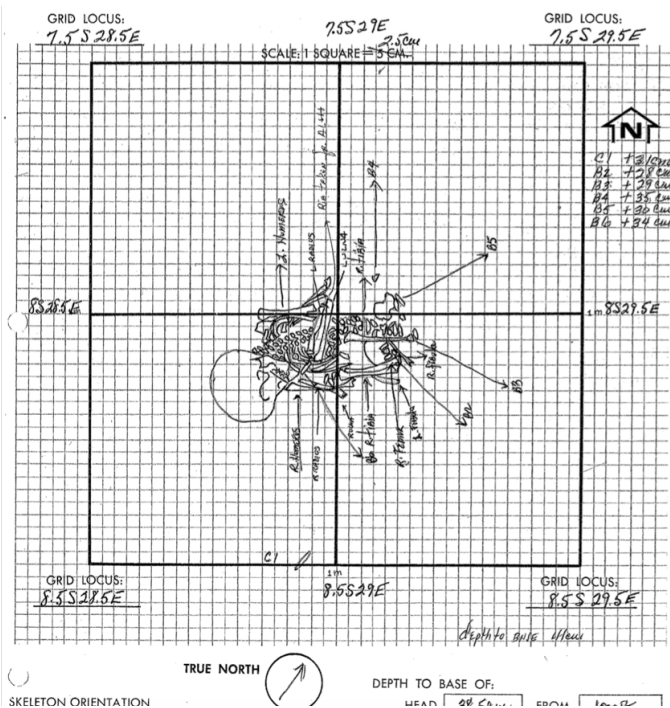
The remains of HN 46-47 S2-10 (Fig. 6) were scattered throughout seven different levels with no articulation between any elements except for the right scapula and right clavicle. An additional frontal bone was recovered from the box while most of the elements of the hands and feet were missing, indicating that some form of processing had occurred. The top slab that had been covering the box had caved in at some point after burial, which likely caused at least some of the damage to the cranium that resulted in 55 fragments. One limestone slab at the base of the box was burned on both sides, though there was no evidence of burnt earth in the vicinity, suggesting that it had been burned at some point prior to its incorporation into the box. A fragment of a Mississippian vessel was recovered and had been incorporated into the structure of the east end of the box. This fragment is believed to be a part of the same vessel associated with HN 46-47 S2-13 (Conner, 1984).

According to the results, HN 46-47 S2-10 was estimated to be 2.5-6 months old at time of death. The Chronology of the Human Dentition chart yielded an age range of 2.5-3 months based on the enamel completion of Ldi_1 and Rdi_1 which occurs at 2.5 months, and the lack of enamel completion of Ldi_2 which occurs at 3 months. The Smith (1991) method was unable to be used because the lower permanent molars were not present. The Ubelaker (1978) method provided an age estimate of 3-9 months based on the formation of Rdm_2 . The long bone diaphyseal lengths yielded consistent results, with estimates of newborn to 6 months. The lower bound was based on the enamel completion of Ldi_1 and Rdi_1 because dental calcification is considered to be one of the best means to estimate age at death (Ubelaker, 1987). Because the long bone diaphyseal lengths yielded consistent results that supported the estimates from the dental data, the upper bound of 6 months was chosen as the upper bound.

The frontal, left and right parietal, and occipital bones were highly fragmented postmortem as evidenced by the jagged edges, which rendered taphonomic analysis of these elements difficult due to issues with bone identification. Nevertheless, the identifiable fragments of the frontal and occipital were able to be definitively scored as Stage 0, with no evidence of taphonomic activity. Unlike the cranial elements, the long bones did not display evidence of postmortem damage. All aspects of the left femur were scored as Stage 0 with no evidence of pitting or other taphonomic activity on any aspects, though minimal white staining was also present and the distal aspect had been damaged postmortem. All aspects of the right femur were also scored as Stage 0, with complete metaphyses, and no evidence of taphonomic activity or postmortem breakage. All aspects of the left and right tibia were also scored as Stage 0, though there was very slight root etching on the posterior aspect of the diaphyses.

The incomplete articulation and excellent preservation of HN 46-47 S2-10 indicates that this child had also received a bundle burial and had then been interred after the soft tissues had disintegrated, as seen with HN 46-47 S2-12 and HN 46-47 S2-13.

Though it is possible that this individual had experienced nutritional stress, trauma, and/or disease in his or her lifetime, there are no skeletally visible indicators of any pathological condition present on any element.



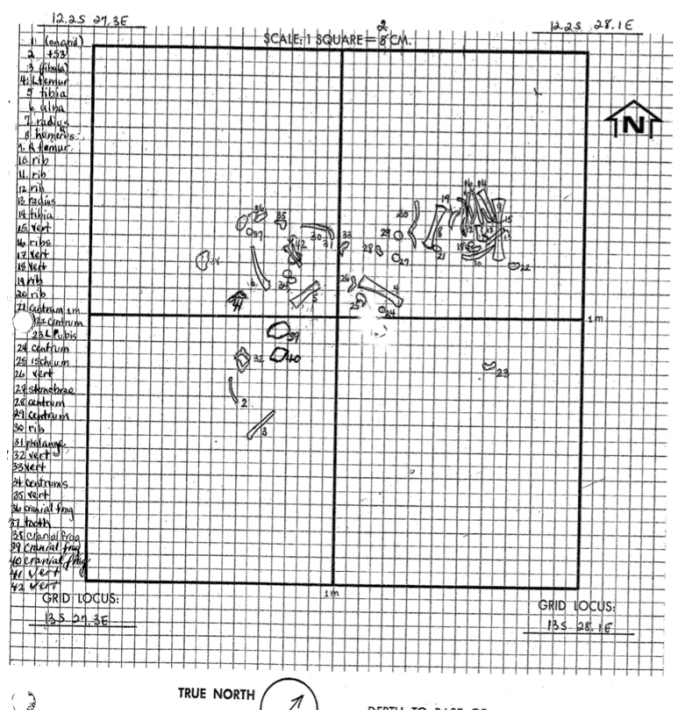


Fig. 6. HN 46-47 H2-10 in its burial context (Johnson, 1974, unpublished data).

Conclusion

The excavations of HN-46 and HN-47 yielded significant finds that provide important information on differential mortuary treatment during the Late Woodland/Mississippian period. Based on dental and skeletal analyses, all individuals were estimated to be under 3 years of age at time of death, while two of the three individuals exhibited evidence of unspecified pathological conditions. The incomplete articulation and minimal taphonomic activity observed on the remains of all individuals suggests that they were protected from the elements during the decomposition process before interment in the stone box graves.

It is possible that the children's ages, illnesses, and/or ethnicity played roles in the ways in which they were interred. Given the proximity of Helton to Cahokia, it is likely that this local community navigated ethnic, religious, social, political, and economic identities through selective Mississippianization while retaining certain local traditions, such as the construction of stone box graves. The mortuary situation at Helton is ambiguous, rendering it nearly impossible to make any definitive claims regarding the significance of the stone box graves, the children interred within them, or the ways in which the local community viewed itself in relation to Mississippian peoples throughout the rise and fall of Cahokia. What can be stated with certainty is that the burial of these children was a significant event (or events) that the community deemed worthy enough to enshrine in stone that would remain untouched for the next 800 years.

References

- Alt SM. 2001. Cahokian change and the authority of tradition. In: T. R. Pauketat, editor. *The archaeology of traditions: agency and history before and after Columbus*. Gainesville: University Press of Florida. p. 141-156.
- Behrensmeyer AK. 1978. Taphonomic and ecologic information from bone weathering. *Paleobiology* 4,2:150-162.
- Buikstra JE. 2016. *Human osteology lab manual*. Kampsville, IL: Arizona State University Field School.

- Buikstra JE, Ubelaker DH. 1994. *Standards for data collection from human skeletal remains*. Fayetteville, AR: Arkansas Archaeological Survey.
- Brown IW. 1981. A study of stone box graves in North America. *Tennessee Anthropologist* 6,1:1-25.
- Conner MD. 1984. Population structure and biological variation in the Late Woodland of Westcentral Illinois [dissertation]. [Chicago, (IL)]: University of Chicago. p. 1-336.
- Delaney-Rivera C. 2004. From edge to frontier: early Mississippian occupation of the Lower Illinois River Valley. *Southeastern Archaeology* 23,1:41-56.
- Farnsworth KB, Emerson TE, Glenn RM. 1991. The Lower Illinois Valley. In: Emerson TE, Lewis RB, editors. *Cahokia and the hinterlands: Middle Mississippian cultures of the Midwest*. Urbana: University of Illinois Press. p. 83-118.
- Lewis ME. 2006. *The bioarchaeology of children: perspectives from biological and forensic anthropology*. Cambridge: Cambridge University Press.
- Logan WG, Kronfeld R. 1933. Development of the human jaws and surrounding structures from birth to the age of fifteen years. *The Journal of the American Dental Association* 20,3(March):379-428.
- Pauketat TR. 1994. *The ascent of chiefs: Cahokia and Mississippian politics in Native North America*. Tuscaloosa: University of Alabama.
- Pauketat TR. 1998. Refiguring the archaeology of greater Cahokia. *Journal of Archaeological Research* 6,1(March):45-89.
- Reynolds AW, Raff JA, Bolnick DA, Cook DC, Kaestle FA. 2015. Ancient DNA from the Schild site in Illinois: implications for the Mississippian transition in the Lower Illinois River Valley. *American Journal of Physical Anthropology* 156,3(March):434-448.
- Roe H. 2011. Stone box grave [image]. *Wikipedia* [Internet]. [cited 2016 Dec 16]. Available from https://en.wikipedia.org/wiki/Stone_box_grave.
- Scheuer L, Black S. 2004. *The juvenile skeleton*. London: Elsevier Ltd.
- Smith BH. 1991. Standards of human tooth formation and dental age assessment. In: Kelley MA, Larsen CS, editors. *Advances in dental anthropology*. New York: Wiley-Liss. p. 143-168.
- Trancho GJ. Inactive cribra orbitalia in a juvenile individual from Copper Age Spain [Image]. *Zoología y Antropología Física* [Internet]. [cited 2017 April 11]. Available from <http://pendientedemigracion.ucm.es/info/antropo/trancho/publicar.htm>.
- Ubelaker DH. 1978. Estimating sex, stature, and age. In: Ubelaker DH. *Human skeletal remains: excavation, analysis, interpretation*. Chicago: Aldine Publishing Company. p. 46-48.
- Ubelaker DH. 1987. Estimating age at death from immature human skeletons: an overview. *Journal of Forensic Sciences* 32,5(September):1254-1263.
- Waldron T. 2009. *Cambridge manuals in archaeology: paleopathology*. Cambridge: Cambridge University Press.

Extraterrestrial Osteology: How Microgravity Affects Human Bone

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More than a century ago, Julius Wolff hypothesized that mechanical loading determined the form and function of bone (Ruff *et al.*, 2006). Since then, “Wolff’s law,” as his hypothesis has come to be known, has been extensively debated and challenged, but as Ruff *et al.* (2006) argued, the underlying process Wolff articulated has stood the test of time even if the colloquially named concept has not (Ruff *et al.*, 2006). Ruff *et al.* (2006), however, argue that the term “bone functional adaptation”—which they trace to an 1881 article by zoologist Wilhelm Roux—rather than Wolff’s controversial “law” might be a more appropriate way of describing the process Wolff initially tried to describe: the feedback loop between applied strain and bone remodeling (Ruff *et al.*, 2006). Essentially, there is an optimum level of strain. If either too much or too little strain is applied, excessive bone deposition or loss (respectively) occurs (Ruff *et al.*, 2006). Within this conceptual framework, it might be tempting to focus on how additional mechanical loading leads to more bone deposition within certain segments of the skeletal system. One might look, for example, at how runners’, manual laborers’ or weight lifters’ skeletons are affected by the abnormal patterns of added strain that characterize their daily lives. This paper focuses on roughly the opposite. It investigates how the mechanical *unloading* that occurs during spaceflight affects human bone. It is nonetheless helpful throughout the present discussion to think about how much ink, thought, and research has been spent pondering the effects of weight and weightlessness on the human skeleton.

By surveying recent research, this paper aims to provide an overview of the osteopenic and osteoporotic effects the microgravity environment has on human bone. After developing an understanding of those effects and a sense of the biochemical means by which they are produced, it surveys some of the leading countermeasures being researched and deployed to combat those effects. In closing, I will briefly consider how this research has implications for NASA’s plans to send humans to Mars.

Microgravity conditions lead to a progressing state of bone demineralization. As weightlessness disrupts bone tissue’s natural equilibrium over time, osteopenic effects start to set in (Smith and Heer, 2002). Left unmitigated, that disruption can lead to disuse osteoporosis, i.e. secondary osteoporosis, or at least a condition that resembles it (Nagaraja and Risin, 2013). Nagaraja and Risin (2013) define osteoporosis as a condition in which there is both a loss of bone mass and changes in bone microarchitectures. This article explores disuse osteopenia and osteoporosis as a continuum of bone loss in which the osteoporotic condition is the extreme case or end result.

There are four primary modalities within which researchers study spaceflight osteopenia. Research takes place both on earth and in space, but extraterrestrial research facilities are relatively new. As such, contemporary understandings of spaceflight-induced osteopenia emerge from a combination of data from both simulated and real microgravity. Space is the ideal environment in which to study bone loss, as it can be monitored in real time and also because the full-body reaction to microgravity—of which bone loss is only one part—can be observed in tandem (Grimm *et al.*, 2016). As Locher *et al.* (2011) note, however, the limited number of available test subjects makes establishing patterns across large data pools difficult. Although space agencies are already

planning a Mars mission, astronauts have only been spending extended periods of time in space since the International Space Station (ISS) was made habitable 16 years ago (Hitt, 2011). An effective and widely used Earth-based alternative is called the “bed rest model,” in which subjects remain head-down in bed at a six-degree tilt for weeks or months on end in order to simulate microgravity conditions (Bickle *et al.*, 1997; Nagaraja and Risin, 2013). A similar research model consists of elevating an animal’s hind limbs in order to simulate fluid shifts and induce bone deconditioning (Nagaraja and Risin, 2013; Canciani *et al.*, 2015). Both models have provided the field with useful data. New in vitro studies are becoming a popular mode of analysis as well. In these studies researchers use clinostats and particle suspension systems in order to disorient bone cells’ gravitational sensors, thereby simulating cellular activity under microgravity conditions (Nabavi *et al.*, 2011; Nagaraja and Risin, 2013; Reggiu and Cancedda, 2015). From clinostats on earth to data recorded at the ISS—and everywhere in between—the scientific community is in the process of constructing a body of knowledge around bone loss due to microgravity.

Bone loss does not occur evenly throughout the skeleton during spaceflight. Researchers have repeatedly shown that spaceflight osteopenia occurs most noticeably in the bone tissue that is placed under significant load on Earth, such as the femoral neck, the pelvis, the lumbar vertebrae, and the calcaneus (Bickle *et al.*, 1997; Nagaraja and Risin, 2013; Canciani *et al.*, 2015; Zhang *et al.*, 2015). Generally, bone loss under microgravity does not reach a level constituting the classification of osteoporosis, but it does in some cases (Ogonov and Bogomolov, 2011). As Nagaraja and Risin (2013) show, however, regardless of the skeletal area in which osteopenic effects sets in during spaceflight, osteoporosity increases more rapidly than aging-induced bone loss.

The osteopenic effect microgravity produces in bone results from the way bone homeostasis becomes disturbed in space. As we know, adult bone forms and is reabsorbed at a relatively stable rate except in cases of remodeling or pathology. On Earth, there is equilibrium among the internal mechanisms that reproduce bone tissue. But in space, osteoclasts, which reabsorb bone matrix, become overactive (Nabavi *et al.*, 2011; Grimm *et al.*, 2016). They overtake the osteoblasts, which methodically lay down new bone matrix (Nabavi *et al.*, 2011). This disturbance in bone’s natural equilibrium leads over time to a net loss of bone, because resorption (osteoclast activity) exceeds production (osteoblast activity) (Nabavi *et al.*, 2011). The exact mechanism by which bone senses loading at the cellular level is subject to debate, but we know that bone does sense changes in load and strain (Bickle *et al.*, 1997). This negative economy of cellular activity does not affect all bone evenly, though, as both Zhang *et al.* (2015) and Grimm *et al.* (2016) have shown. Its intensity within a given bone is dependent upon the weight under which that bone is normally placed at Earth’s gravity. In essence, though, despite how osteopenic effects are present in deferent degrees across the human skeleton, weightlessness affects bone cells all the way down from their cellular adhesion sites to their nuclei (Nabavi *et al.*, 2011). Microgravity disturbs the delicately orchestrated cycles of activity between osteoblasts, osteocytes, and osteoclasts (Grimm *et al.*, 2016). This metabolic disturbance, by which bone is progressively

demineralized, leads to osteopenia and, in some cases, eventually to disuse osteoporosis (Nabavi *et al.*, 2011).

The disruptions in bone metabolism microgravity triggers lead to a range of negative effects, such as calcium deficiency and even renal stones (Smith and Heer, 2002). Scott Smith has consistently demonstrated that degenerative process over 12 years of research. He has shown how weightlessness leads to functional adaptations in bone endocrine regulation, and how these adaptations reflect decreased calcium uptake due to accelerated bone resorption (Smith and Heer, 2002; Smith *et al.*, 2015). Smith *et al.* (2012) discuss how evidence from both bed rest and spaceflight studies suggests that unloading stimulates bone decalcification, which suppresses parathyroid hormone release, and moreover how the sum total of this biochemical disruption leads to net bone loss. Microgravity's osteopenic effects are part of a larger biochemical disruption in bone and calcium metabolism.

The rate at which osteopenic effects set in is important to consider, especially given that several governmental and corporate space agencies are envisioning interplanetary flight. Such missions would require humans to stay in space for between one and two years (NASA's Journey to Mars... [updated 2014]). In general, researchers have determined that spaceflight osteopenia develops quickly, with demineralization occurring at a rate of one to two percent per month (Oganov and Bogomolov, 2011; Nagaraja and Risin, 2013). They show, however, that spaceflights no longer than seven months generally do not lead to enough bone loss to constitute a classification of osteoporosis, but also that such long-term missions can and produce significant yet generally reversible osteopenia (Oganov and Bogomolov, 2011). As shown above, Smith *et al.* (2012) suggest that the biochemical reactions to microgravity set in rapidly even if noticeable bone loss does not occur as quickly.

In sum, by disrupting bone homeostasis, microgravity environments produce osteopenic effects that over a period of several months can lead toward secondary osteoporosis. Gliding through space, astronauts may not personally notice what is happening to their bodies. After all, it is not as though they could feel any lighter! But because they have to engage in demanding physical activities, such as repairing spacecraft and moving objects around, the risk of fracture is a serious concern for spaceflight missions (Space Bones... [updated 2011]). Moreover, if fractures or bone tissue injuries do not occur in space, there is a dramatic risk of finding an unexpectedly hard landing upon returning to Earth's gravity. For the field of space exploration, the progressing demineralization of bone in microgravity is a major concern to which researchers are determined to find solutions.

For as long as the scientific community has been aware of the way bone reacts to microgravity it has been searching for ways of mitigating those effects. Solutions such as free-weight training, exercise equipment (e.g., treadmills), drugs, and nutritional supplements have been—and are being—researched and deployed (Nagaraja and Risin, 2013). There are unique challenges facing each of these proposed regimes, though, resulting from the restrictions space places on research and engineering solutions (Loehr *et al.*, 2011). For example, as Loehr *et al.* (2011) show, in order to justify deploying a piece of exercise equipment to the ISS, it must not vibrate; it can require only minimal external power; lastly, it must be able to be easily maintained over a 15-year period. Whichever countermeasures prove most effective in the end, clearly their efficacy depends upon more than just their capacity to offset bone loss.

Load-bearing suits that exert pressure upon the skeletal (and muscular) system are one kind of countermeasure that has been used to offset disuse osteopenia. Russian cosmonauts staying at the ISS have been donning “Pingvin” (i.e., “penguin”) suits since the 1970s (Waldie and Newman, 2010). The simple fact that the suit's creators named it after a penguin hints at how uncomfortable it is; it evokes the suit's non-human parameters. Waldie and Newman (2010) report that cosmonauts

regularly complain of overheating issues and extreme discomfort when wearing the suit. Moreover, despite its widespread use, ground-based studies have not shown that it effectively mitigates bone loss (Waldie and Newman, 2010). But we cannot write off gravity loading suits entirely. Taking the Pingvin suit's faults as a starting point, Waldie and Newman (2010) developed an alternative or at least the concept for an alternative. They call it the “Gravity Loading Countermeasure Skinsuit,” (or GLCS), which they have shown to be a much more comfortable and streamlined solution during parabolic test-flights (Waldie and Newman, 2010). But whether or not the GLCS is truly capable of offsetting osteopenic effects, and for how long it could do so, remains to be seen under actual microgravity conditions.

Load-bearing suits may hold promise after future research, but they are not the only solution. Various forms of exercise equipment show potential. For example, NASA has been showcasing a new exercise machine called the Advanced Resistive Exercise Device (ARED), which is able to simulate controlled free-weight training (Advanced Resistive Exercise Device... [updated 2015]). Researchers have shown that the ARED is an effective means for mitigating both bone loss and muscle atrophy, both of which are components of the generalized deconditioning of the body in space (Loehr *et al.*, 2011; Advanced Resistive Exercise Device... [updated 2015]). Analysis also indicates that treadmills could be a useful countermeasure to disuse osteopenia (De Witt and Ploutz-Snyder, 2014). De Witt and Ploutz-Snyder (2014) demonstrate how, by simply increasing the operating speeds of treadmills currently on the ISS, mechanical loading would approach Earth gravity levels, which could stabilize or at least slow bone deconditioning. By contrast, both Smith and Heer (2002) and later Nagaraja and Risin (2013) assert that exercise alone has proven to be an ineffective countermeasure to bone loss and that the best solution is to coordinate the use of resistive devices with dietary and pharmacological regimes. But regardless, as Loehr *et al.* (2011) point out, the particular engineering difficulties that space presents will ultimately determine which countermeasures are deemed most effective and most practical. There appears to be far more consensus regarding the causes of spaceflight osteopenia than there is concerning how to combat it.

Microgravity causes gradual yet serious effects in bone tissue. By destabilizing bone metabolism, it accelerates bone resorption beyond the level at which new bone can form. Over a period of several months, that instability leads to disuse osteopenia or even secondary osteoporosis. Although these effects are generally reversible, they remain major hindrances to space programs' plans to penetrate ever deeper into space in the coming decades. There is some evidence to suggest that osteopenia could be stabilized during long-term spaceflight, but further research should be conducted to probe that hypothesis (Oganov and Bogomolov, 2011). As we have seen, there are several exercise regimes being researched and deployed that could lessen or at least counteract the effects of weightlessness (Waldie and Newman, 2010; Loehr *et al.*, 2011; Nagaraja and Risin, 2013; De Witt and Ploutz-Snyder, 2014). Still, bone loss remains a major hurdle yet to be overcome. The fact that there is a growing body of research around the issue is a promising sign that effective solutions will be discovered.

As a species, we have been staring into the maw of deep space for millennia, wondering what—or *who*—is in that seemingly endless elsewhere. Very soon, we may be traveling into deep space instead of simply gazing. In a 2010 bipartisan outline, NASA laid out plans to send humans to an asteroid by 2025 and to Mars during the 2030s (NASA's Journey to Mars... [updated 2014]). NASA is shifting its focus from low Earth orbit missions toward deep space. Some researchers have concluded that there is no concrete way of establishing that a deep space mission of one-to-two years would produce osteoporosis beyond the level that has been observed on shorter missions (Oganov

and Bogomolov, 2011). But nonetheless, we will need to be sure before we send humans tumbling into deep space. Moreover, considering that Mars' gravity is 38 percent of Earth's, it is important that we look into not only how microgravity (i.e., zero gravity) affects bone in relation to Earth's gravity but also how sustained yet partial gravity will affect the skeleton (Smith and Heer, 2002). Perhaps after future scientific discovery, we could develop ways of stabilizing bone metabolism on the red planet even if doing so in zero gravity remains problematic. That is something only additional investigation efforts could determine. Bone loss is not the most significant problem for deep space missions. It is nevertheless a challenge researchers and engineers have yet to overcome.

References

- Advanced Resistive Exercise Device (ARED) [Internet]. [updated 2015 June 18]. Washington, D.C.: National Space Agency (NASA). [cited 2016 Apr 23]. Available from: http://www.nasa.gov/mission_pages/station/research/experiments/1001.html
- Bikle DD, Halloran BP, Morey-Holton E. 1997. Space flight and the skeleton: lessons for the earthbound. *Endocrinologist* 7,1:10-22.
- Canciani B, Ruggiu A, Giuliani A, Panetta D, Marozzi K, Tripodi M, Salvadori PA, Cilli M, Ohira Y, Cancedda R, Tavella S. 2015. Effects of long time exposure to simulated micro- and hypergravity on skeletal architecture. *Journal of the Mechanical Behavior of Biomedical Materials* 51,1(November):1-12.
- De Witt JK, Ploutz-Snyder LL. 2014. Ground reaction forces during treadmill running in microgravity. *Journal of Biomechanics* 47,10(July):2339-2347.
- Grimm D, Grosse J, Wehland M, Mann V, Reseland JE, Sundaresan A, Corydon TJ. 2016. The impact of microgravity on bone in humans. *Bone* 87(June):44-56.
- Hitt, D. 2011. What is the International Space Station?. *NASA Knows! (Grades K-4)* [Internet]. [cited 2016 Apr 23]; 11:30. Available from: <http://www.nasa.gov/audience/forstudents/k-4/stories/nasa-knows/what-is-the-iss-k-4.html>
- Loehr JA, Lee SMC, English KL, Sibonga J, Smith SM, Spiering BA, Hagan RD. 2011. Musculoskeletal adaptations to training with the advanced resistive exercise device. *Medicine and Science in Sports and Exercise* 43,1(January):146-156.
- Nabavi N, Khandani A, Camirand A, Harrison RE. 2011. Effects of microgravity on osteoclast bone resorption and osteoblast cytoskeletal organization and adhesion. *Bone* 49,5(November):965-974.
- Nagaraja MP, Risin D. 2013. The current state of bone loss research: data from spaceflight and microgravity simulators. *Journal of Cellular Biochemistry* 114,5(May):1001-1008.
- NASA's Journey to Mars [Internet]. [updated 2014 Sep 14]. Washington, D.C.: National Space Agency (NASA); [cited 2016 Apr 23]. Available from: <http://www.nasa.gov/content/nasas-journey-to-mars>
- Oganov VS, Bogomolov VV. 2011. The human skeletal system in weightlessness: a review of research data, hypotheses, and the possibility of predicting the state in long-term (interplanetary) missions. *Human Physiology* 37,7(December):768-776.
- Ruff C, Holt B, Trinkaus E. 2006. Who's afraid of the big bad Wolff?: "Wolff's Law" and bone functional adaptation. *American Journal of Physical Anthropology* 129,4(April):484-498.
- Ruggiu A, Cancedda R. 2015. Bone mechanobiology: gravity and tissue engineering: effects and insights. *Journal of Tissue Engineering and Regenerative Medicine* 9,12(December):1339-1351.
- Smith SM, Heer M. 2002. Calcium and Bone Metabolism During Spaceflight. *Nutrition* 18,10(October):849-852.
- Smith SM, Heer M, Shackelford LC, Sibonga JD, Spatz J, Pietrzyk RA, Hudson EK, Zwart SR. 2015. Bone metabolism and renal stone risk during International Space Station missions. *Bone* 81(December):712-720.
- Smith SM, McCoy T, Gazda D, Morgan JLL, Heer M, Zwart SR. 2012. Space flight calcium: implications for astronaut health, spacecraft operations, and Earth. *Nutrients* 4,12(December):2047-2068.
- Space Bones [Internet]. [updated 2011 Apr 6]. National Space Agency (NASA); [cited 2016 Apr 23]. Available from: http://science.nasa.gov/science-news/science-at-nasa/2001/astor1oct_1/
- Waldie JM, Newman DJ. 2011. A gravity loading countermeasure skinsuit. *Acta Astronautica* 68,7-8:722-730.
- Zhang J, Li J, Xu H, Yang P, Xie L, Qian A, Zhao Y, Shang P. 2015. Responses of bone cells to microgravity: ground-based research. *Microgravity Science and Technology* 27,6(July):455-464.



